

DATA SCIENCE

MATHEMATICS BEHIND DATA SCIENCE & MACHINE LEARNING.

✓ Mathematics for Data Science and ML Coverage:

- Probability and Statistics.
 - Calculus
 - Linear Algebra.
 - Discrete Mathematics.
 - Information Theory.
 - Stochastic Processes.
- } 3. most Used.

Probability & Statistics.

Statistics

- Is a science which studies principles and methods applied in collecting, presenting, analysing and interpreting the numerical data of any data field of inquiry.

Importance of Statistics.

1. Data grouping
2. Algorithm performance and evaluation.
3. Data analysis.
4. Machine Learning. & AI, etc.

Presentation of Data

Terminologies.

1) Discrete Variable

- Data can be obtained using the following two methods:-

1. Counting - using the counting a researcher wants to know the number of students in statistics class, he can do that by counting the students.
2. Measurements - using measurements we use different instruments depending on what is measured. Example if I want to know a temperature of a given room we use a thermometer.

- When data are obtained by counting and measuring, and only whole numbers are possible, the data are called Discrete.

- The opposite of Discrete is called continuous.

Examples of Discrete data

1. 0, 1, 2, 3, 4, 5 → Number of cars in a parking lot.

NB ⇒ Whole numbers Only.
They are countable not measurable.

ii) Continuous Variable

- Data which are measured and taken data any value within a setup limits are called continuous.

Examples of Continuous Data

1.5, 2.3, 2.75, 3.1 → Heights of Students (in meters).

NB ⇒ Fractions & Decimals Only.
They are measurable, not countable.

iii) A Set

- Is a group of data.

- An individual value within the set is called a member of a set.

ie The masses of 5 students in a class are 42.3kg, 57.4kg, 60.7kg and 72.5kg

Set
member of a set

{42.3, 57.4, 60.7, 72.5} = A Set

- A Set which contains all members is called a population.

{42.3, 57.4, 60.7, 72.5, 77.2}

Population
complete member sets

- Some members selected at random from a population are called a sample.

{42.3, 57.4, 60.7, 72.5, 77.2}

Sample

All reg. registrations numbers in Kenya, they form a population, suppose 20 registration numbers are put in consideration class, this is called a sample. Suppose a researcher wants to know the number of Computer Science students, he picks a few from university in Kenya to act as a sample. This information for all will going to give us the attitude of all students in Kenya studying Computer Science.

iv) Frequency - The number of times the values of a member occurs in a set is called frequency of that member.
Example.

{2, 3, 5, 4, 7, 5, 6, 2, 8}

2 - twice	} occurring frequently.
3 - once	
5 - twice	
4 - once	
7 - once	
6 - once	
8 - once	

v) Relative Frequency - is the ratio of frequency of a member to the total frequency of all members.

$$R.F = \frac{\text{Frequency of a member}}{\text{Total Frequency of all members.}}$$

Example - Given a Set:

{10, 3, 5, 7, 2, 5, 10, 6, 4}

Find the frequency of each member and Relative Frequency (R.F):

Frequency of each member.

- 2 - once
- 3 - once
- 4 - once
- 5 - twice
- 6 - once
- 7 - once
- 10 - thrice

n) R.F

$$R.F = \frac{f \text{ of each member}}{\text{Total Frequency}}$$

- 2 - $\frac{1}{10}$ or $1:10$
- 3 - $\frac{1}{10}$ or $1:10$
- 4 - $\frac{1}{10}$ or $1:10$
- 5 - $\frac{2}{10}$ or $2:10$
- 6 - $\frac{1}{10}$ or $1:10$
- 7 - $\frac{1}{10}$ or $1:10$
- 10 - $\frac{3}{10}$ or $3:10$

Percentage Relative Frequency
This refers to the Relative Frequency $\times 100\%$

$$R.F\% = R.F \times 100\%$$

Example:

Using the previous example question, Find the Percentage R.F?

- 2 - $\frac{1}{10} \times 100\% = 10\%$
- 3 - $\frac{1}{10} \times 100\% = 10\%$
- 4 - $\frac{1}{10} \times 100\% = 10\%$
- 5 - $\frac{2}{10} \times 100\% = 20\%$
- 6 - $\frac{1}{10} \times 100\% = 10\%$
- 7 - $\frac{1}{10} \times 100\% = 10\%$
- 10 - $\frac{3}{10} \times 100\% = 30\%$

Exercise for practice

Data obtained in the topic below state whether they are discrete or continuous data:-

- i) The number of days in which rain falls in a month for each month of the year, this data contains counting Discrete data $\Rightarrow$ because it has trained.
- ii) The mileage traifard by each sales man, of data is Continuous data $\Rightarrow$ because this type of data is measurable (the mileage is measured).
- iii) The time taken by a student from home to school, Discrete data $\Rightarrow$ because we count the number of hours or minutes taken not measured and they are whole numbers.
- iv) The amount of money spent by grade 10 Student to be admitted to a junior school, Discrete data $\Rightarrow$ because counting is present and also the amount are whole numbers.
- v) The amount of petrol produced daily in a certain company, Continuous data $\Rightarrow$ because amount of petrol is measurable (fractions and decimals).
- vi) The number of bottles delivered daily by each of 30 milk men, Discrete data $\Rightarrow$ because the number of bottles are countable.

2. Given the following sets.

Red, blue, green, yellow, red, blue, pink, red, purple }

Calculate the relative frequency of each member calculate the relative percentage of each member.

Red - thrice
Blue - twice
Green - once
Yellow - once
Pink - once
Purple - once.

$$\text{Red} = \frac{3}{8} \times 100\% = 37.5\%$$

$$\text{Blue} = \frac{2}{8} \times 100\% = 25\%$$

$$\text{Green} = \frac{1}{8} \times 100\% = 12.5\%$$

$$\text{Yellow} = \frac{1}{8} \times 100\% = 12.5\%$$

$$\text{Pink} = \frac{1}{8} \times 100\% = 12.5\%$$

$$\text{Purple} = \frac{1}{8} \times 100\% = 12.5\%$$

3. Calculate the same:

{5, 10, 15, 20, 10, 20, 20, 35, 70, 40}

<u>R.F</u>		<u>R.F</u>	
5 $\Rightarrow$ once	$\frac{1}{10} \times 100\% = 10\%$	1 $\times 100\% = 10\%$	
10 $\Rightarrow$ twice	$\frac{2}{10} \times 100\% = 20\%$	10 $\times 100\% = 10\%$	
15 $\Rightarrow$ once	$\frac{1}{10} \times 100\% = 10\%$	70 $\times 100\% = 10\%$	
20 $\Rightarrow$ thrice	$\frac{3}{10} \times 100\% = 30\%$		
35 $\Rightarrow$ once			
70 $\Rightarrow$ once			
40 $\Rightarrow$ once			

TYPES OF DATA

i) Discrete Data also known as ungrouped data.

ii) Continuous Data also known as grouped data.

Ungrouped data is data that is presented in its raw or original form, without being arranged into classes or intervals.

Grouped data is data that is organised into class intervals (groups) with their corresponding frequencies.

Ungrouped data can be presented via, diagrammatically by use of pictogram, horizontal bar charts, vertical bar charts:-

- i) Pictogram, } Ungrouped Data
- ii) Horizontal Bar Charts.
- iii) Vertical Bar Charts.

i) Pictogram.

- It uses symbols to represent quantities.

Q.N. The number of television sets provided in a workshop by a technician in 6 months is shown below:

Present this data in a pictogram:-

Months	NO of TV prepared.
Jan	11
Feb	6
Mar	15
Apr	9
May	13
June	8

$N/B \Rightarrow$ We divide by 2 $\rightarrow$ use a scale not necessarily 2 on the amount. This is $\rightarrow 5\frac{1}{2}$ half symbol.

$$\begin{aligned} 11 &= 5\frac{1}{2} \\ 2 &= 3 \\ 2 &= 7\frac{1}{2} \\ 15 &= 4\frac{1}{2} \\ 2 &= 4\frac{1}{2} \\ 13 &= 6\frac{1}{2} \\ 2 &= 4 \\ 8 &= 4 \end{aligned}$$

Months	NO of TV sets.
Jan	□□□□□
Feb	□□□□
Mar	□□□□□□□□
April	□□□□□□□
May	□□□□□□□□
June	□□□□□

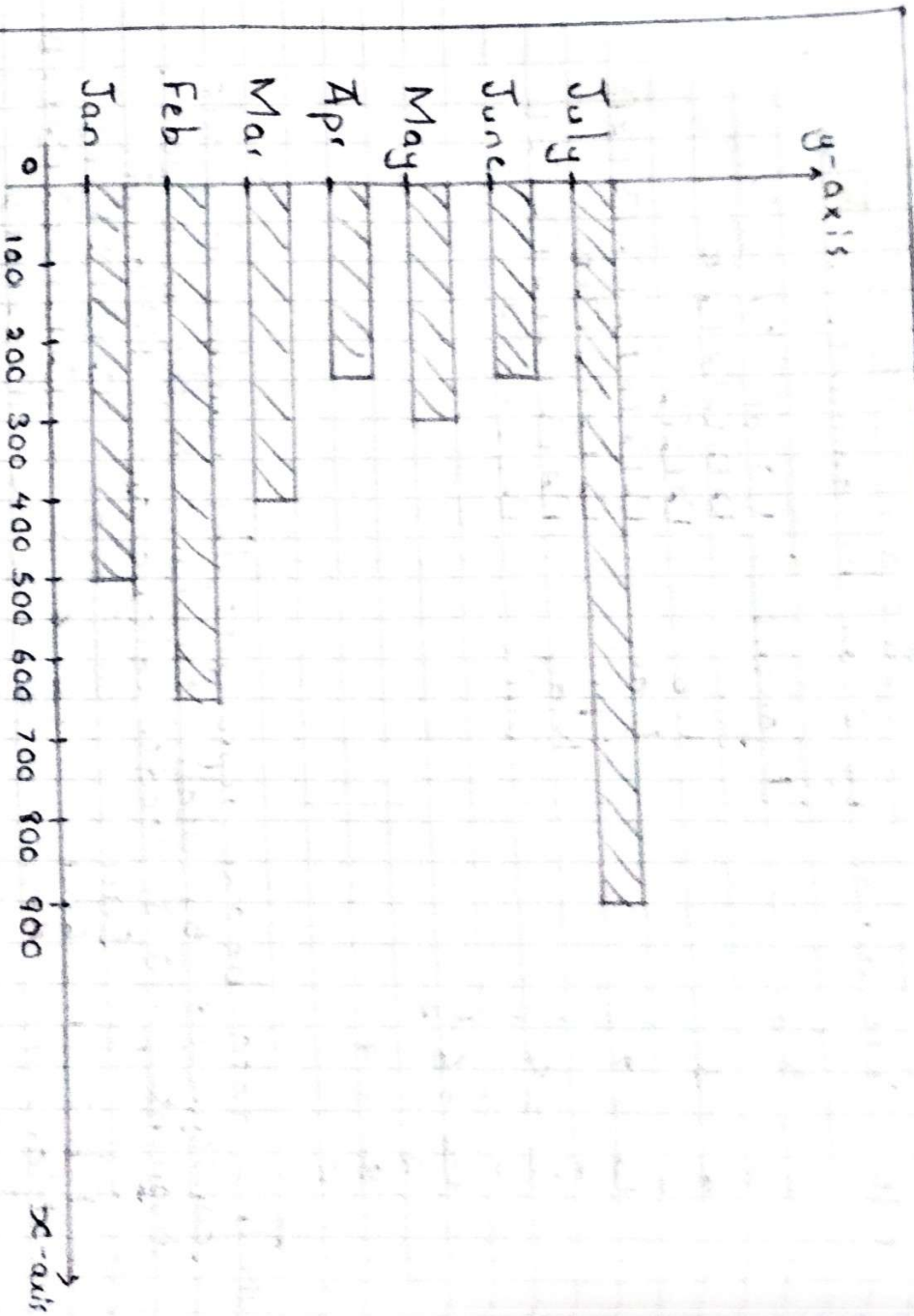
Q4) The data below represents the number of seeds constructed by Uthali turndures in the year 2024.

Months	NO of seeds.	Calculations.	Pictogram.
Jan	500	$500 \div 100 = 5$	□□□□□
Feb	650	$650 \div 100 = 6.5$	□□□□□□□
Mar	400	$400 \div 100 = 4$	□□□□
June	250	$250 \div 100 = 2.5$	□□□
July	900	$900 \div 100 = 9$	□□□□□□□□
May	300	$300 \div 100 = 3$	□□□
April	250	$250 \div 100 = 2.5$	□□□

ii) Horizontal Bar Chart

Is a type of chart used to display and compare different categories of data using horizontal rectangular bars. In this chart, the categories are usually shown on the vertical axis, while the values are shown on the horizontal axis.

On the question of seeds representation represent the same data on a Horizontal Bar Chart.



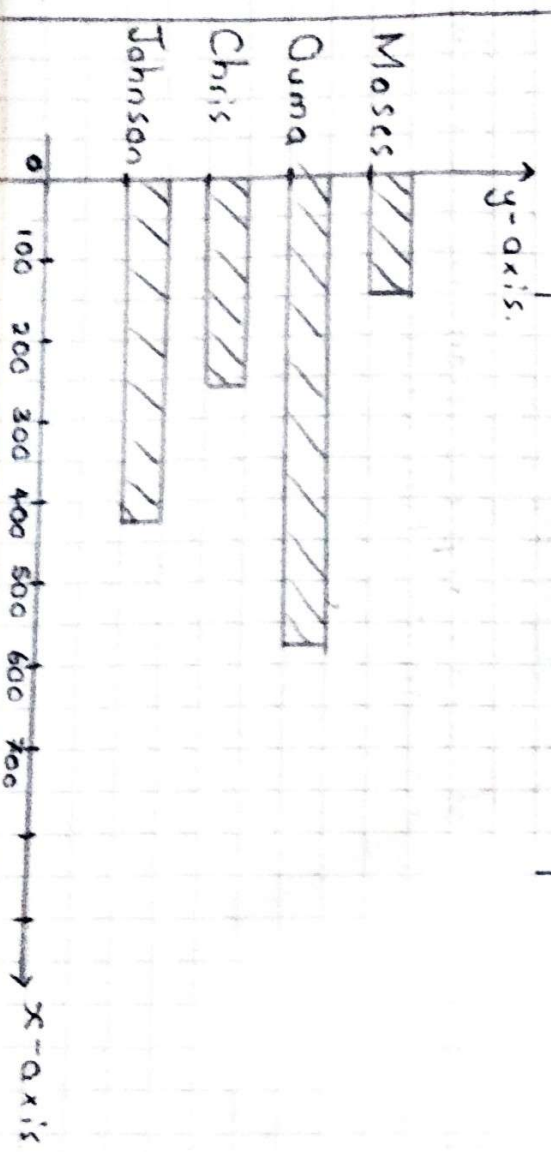
Q4. In this diagram, data are represented by equally spaced horizontally rectangles. Examples, the distance in miles travelled by 4 sales men in a week are shown below.

Diagrammatically

Salesmen

Distance

Johnson 413
Chris 264
Ouma 597
Moses 143

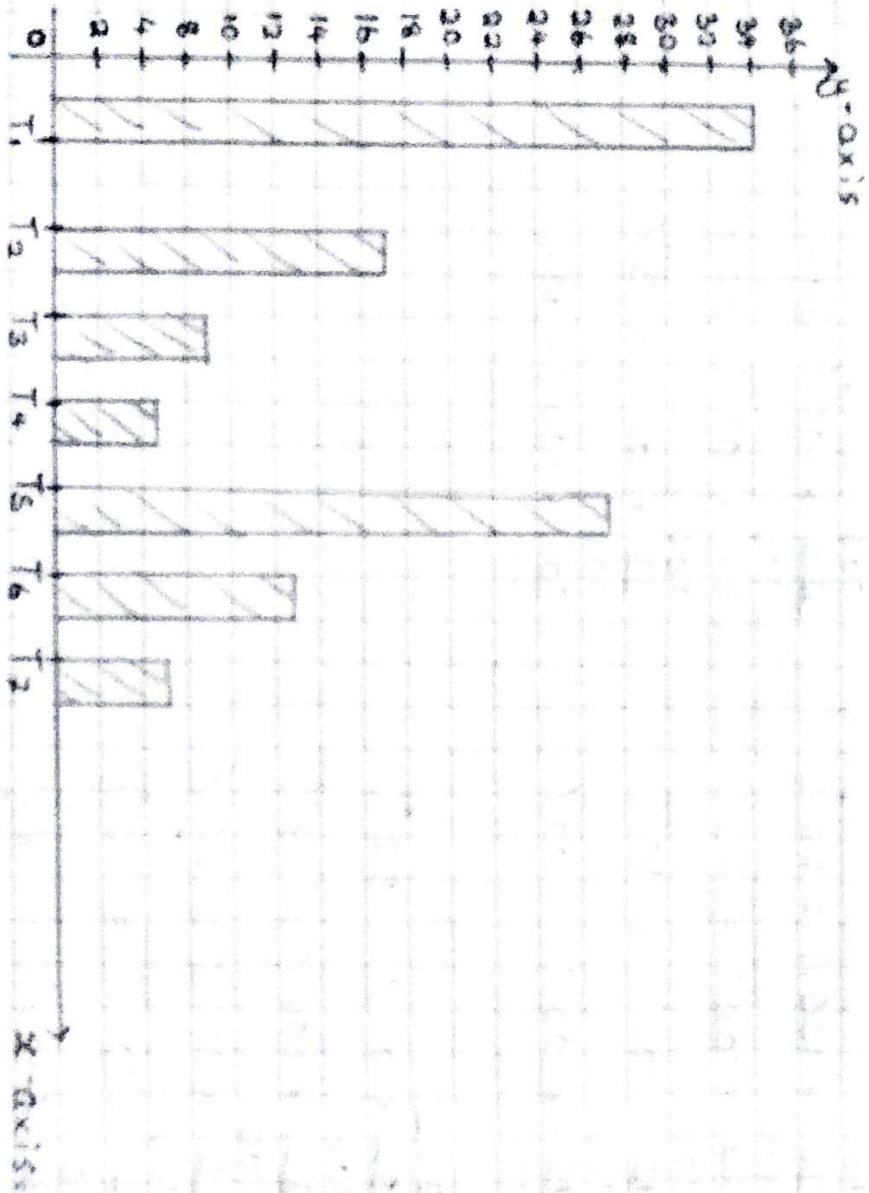


iii) Vertical Bar Chart

Data are represented in equally spaced rectangles vertically in this chart. The values are on the vertical axis and the category are usually on the horizontal axis.

N. The Number of materials in a store in a factory is listed below in a day. Represent the data in a vertical bar chart.

Time	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇
No of materials	34	17	9	5	27	13	9



% Component Bar Chart

It is a type of graph that shows how different parts contribute to a whole using percentages. It helps to display the proportion of each component in relation to the total.

Table below shows the percentage distribution of a company's sales in 4 departments over three different years.

Department	2021	2022	2023
Electronics	30	35	32
Furniture	25	20	23
Clothing	20	22	25
Groceries	15	13	12
Other items	10	10	8

a) Draw a percentage component graph to represent the company's sales distribution for the 3 years.

$$\frac{30}{100} \times 100\% = 30\%$$

$$\frac{25}{100} \times 100\% = 25\%$$

$$\frac{20}{100} \times 100\% = 20\%$$

$$\frac{15}{100} \times 100\% = 15\%$$

$$\frac{10}{100} \times 100\% = 10\%$$

$$\frac{35}{100} \times 100\% = 35\%$$

$$\frac{20}{100} \times 100\% = 20\%$$

$$\frac{22}{100} \times 100\% = 22\%$$

$$\frac{13}{100} \times 100\% = 13\%$$

$$\frac{10}{100} \times 100\% = 10\%$$

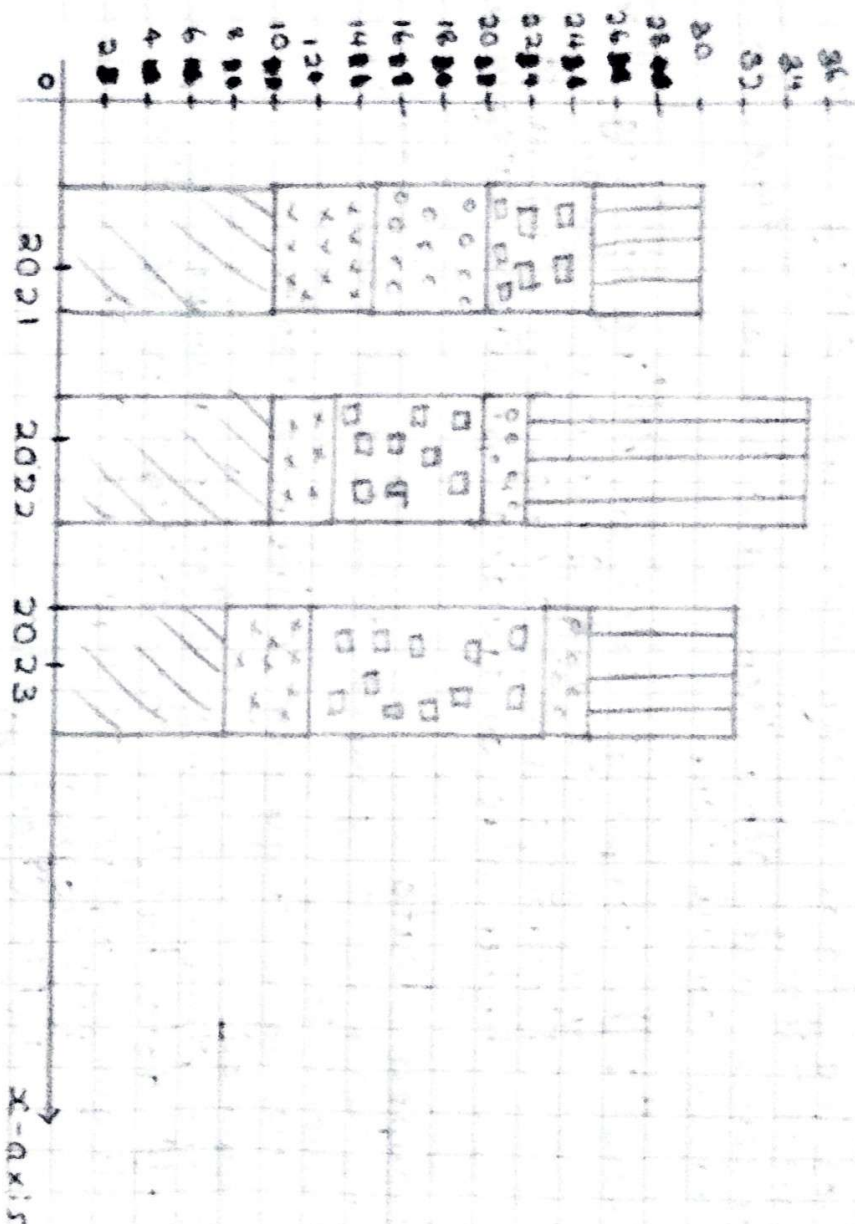
$$\frac{32}{100} \times 100\% = 32\%$$

$$\frac{23}{100} \times 100\% = 23\%$$

$$\frac{25}{100} \times 100\% = 25\%$$

$$\frac{12}{100} \times 100\% = 12\%$$

$$\frac{8}{100} \times 100\% = 8\%$$



Key must be present.

Key	Symbol
Electronics	□
Furniture	□
Clothing	□
Groceries	□
Other Items	□

- b) Identify which department had the highest percentage of sales in 2022?
 - c) Explain the trend in clothing sales from 2021 to 2023.
- The amount of sales of clothing gradually increased in amount of sales made between 2021 to 2023, from 20% to 25% over the 3 years.

- (d) What conclusion can you make from the Company's sales pattern?
- Electronics consistently led in sales, though it dipped slightly in 2023.
 - Furniture fluctuates but remains ~~high~~ around 20%-25%.
 - Clothing shows steady increase, indicating a growing market share.
 - Groceries and other items decrease over time, showing reduced contribution.

Overall: The company's sales are ~~rising~~ shifting slightly from groceries and other items towards Electronics and Clothing, suggesting a focus on higher value or trending products categories.

Grouped Data

We can present grouped data through the following methods :-

- Using a frequency distribution table.

Frequency

Is the number of times a (letter) number is appearing.

- If data has less than 10 items, we use the frequency method to represent data:-

- i) Histogram
- ii) Pie Chart
- iii) Bar Chart

- If data of items is more than 10, we use the following methods to represent it :-

- i) Frequency Distribution Table

Frequency Distribution Table

- Classes: Represents the range of items. To assist in (70-78), counting the members (items), we use a tally diagram.

A (grid) is a table showing classes and their corresponding frequencies.

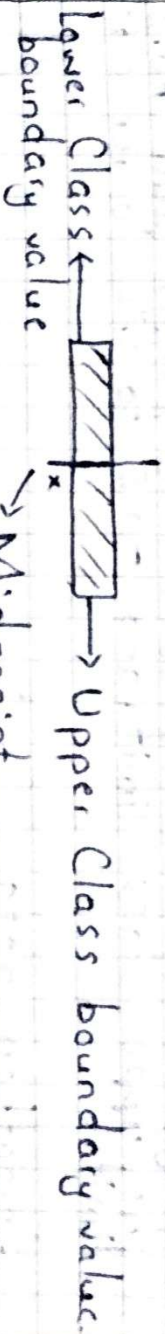
The new set of values, formed by frequency distribution is called a grouped data.

The size range of a class is given by :-

Range = (Upper class boundary value) - (Lower class boundary value)

The Mid-point is obtained by:

Mid-point = $\frac{\text{Upper Class boundary value} + \text{Lower class boundary value}}{2}$



Interval for each Class = $\frac{\text{Range}}{\text{NO of Classes}}$

Exercise

Q1. The Data given below refers to the gain of each of a bunch of 40 transistors. Express correct to the nearest whole number, Form a frequency distribution data showing 7 Classes.

81	86	89	81	84	86	78	81	73	78	77	80	78
80	74	71	81	80	85	76	86	73	85	80	89	85
89	78	79	82	87	82	84	80	84	86	79	78	77

Range = $\frac{\text{Highest}}{89} - \frac{\text{Lowest}}{71}$

= 18

Interval for each class = $\frac{\text{Range}}{\text{NO of Classes}}$

= $\frac{18}{7}$

= 2.5714

= $\frac{3}{7}$

Class	Tally Sheet	Frequency	Mid-p	Class-b
70-72	/	1	71	69.5-72.5
73-75	///	3	74	72.5-75.5
76-78	####	5	77	75.5-78.5
79-81	####	12	80	78.5-81.5
82-84	####	7	83	81.5-84.5
85-87	####	7	86	84.5-87.5
88-90	####	1	89	87.5-90.5

To get a Smoothed data we always add $+0.5$ to the upper class limit and -0.5 to the lower class limit.

Smoothen data

$$\frac{73-72}{2} = 0.5 \text{ or } \frac{1}{2} \text{ so } +0.5 \text{ to get}$$

Smoothen = $\frac{\text{Upper Class Limit} - \text{Lower Class Limit}}{2}$.

Exercise.

The amount of money earned weekly by 40 people daily in a factory, correct to the nearest shilling is shown below:
Draw a frequency distribution having 6 classes having this data.

80	90	70	110	90	90	120
140	90	90	50	100	110	40
80	90	110	80	100	30	110
130	130	80	120	160	110	80
50	100	110	90	110	60	100
70	110	80	100	180	high	

* Prepare a frequency distribution.

- * Class
- * Tally
- * Frequency
- * Mid-point
- * Class-boundary
- * Smoothen the Data

Class	Tally	Sheet	Frequency	Mid-p	Class-boundary
25-50			4	37.5	24.5-50.5
51-76			5	38.5	50.5-76.5
77-102			5	89.5	76.5-102.5
103-128			5	115.5	102.5-128.5
129-154			5	141.5	128.5-154.5
155-180			5	167.5	154.5-180.5

$$\text{Interval} = \frac{\text{Range}}{\text{No. of Class}}$$

$$\text{Range} = 180 - 25 = 155$$

$$I = \frac{155}{6} = 25$$

$$\text{C. boundary} = \frac{51-50}{2} = \frac{1}{2} \text{ or } 0.5$$

Graphs for Grouped Data

ii) Histogram

- Made up of a series of vertical rectangles which are adjacent to each other and are made in proportion to the frequency of the classes.
- When class intervals are equal, the height of the rectangle in a histogram are equal to the frequencies of the classes.
- For Histograms having unequal class intervals, the area must be proportional to the frequency.

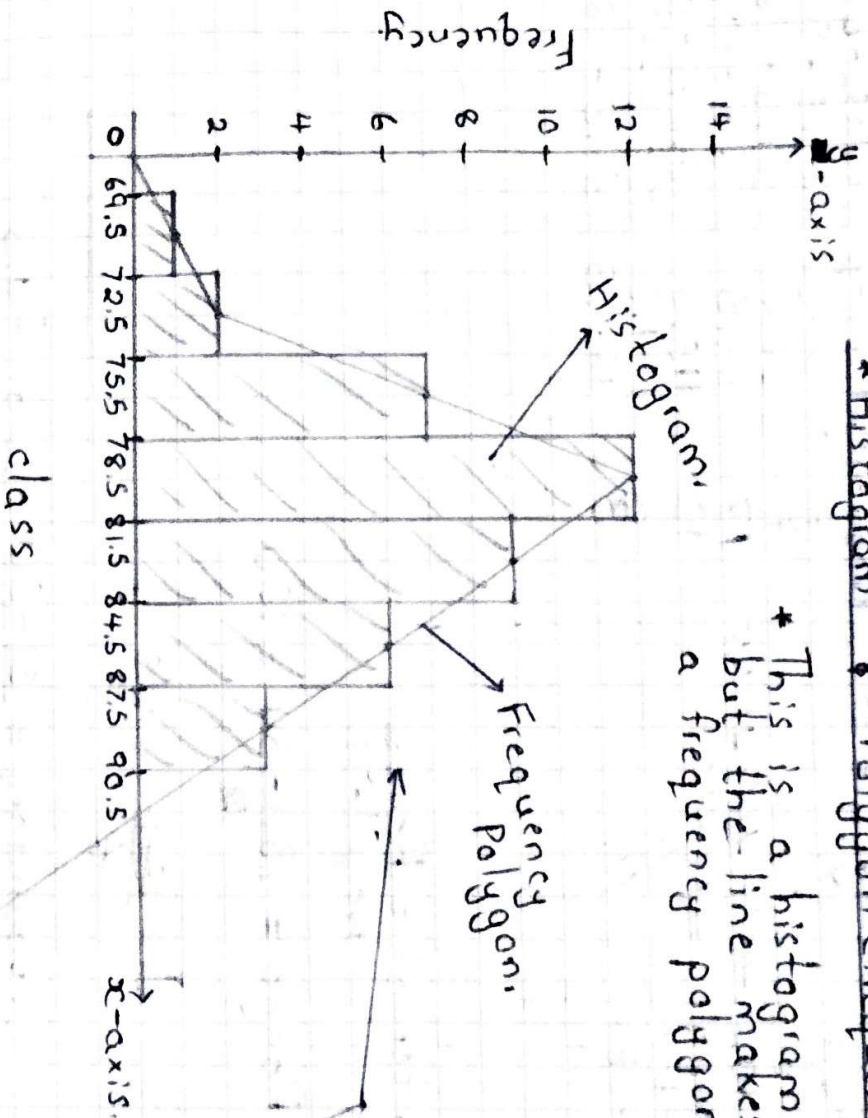
Exercise

Q4. Construct a histogram using the data given below.

Class	Mid-point	Frequency	Class-b	cf
70-72	71	1	69.5-72.5	1
73-75	74	2	72.5-75.5	3
76-78	77	7	75.5-78.5	10
79-81	80	12	78.5-81.5	22
82-84	83	9	81.5-84.5	31
85-87	86	6	84.5-87.5	37
88-90	89	3	87.5-90.5	40

* Histogram & Polygon Frequency

* This is a histogram but the line makes it a frequency polygon.

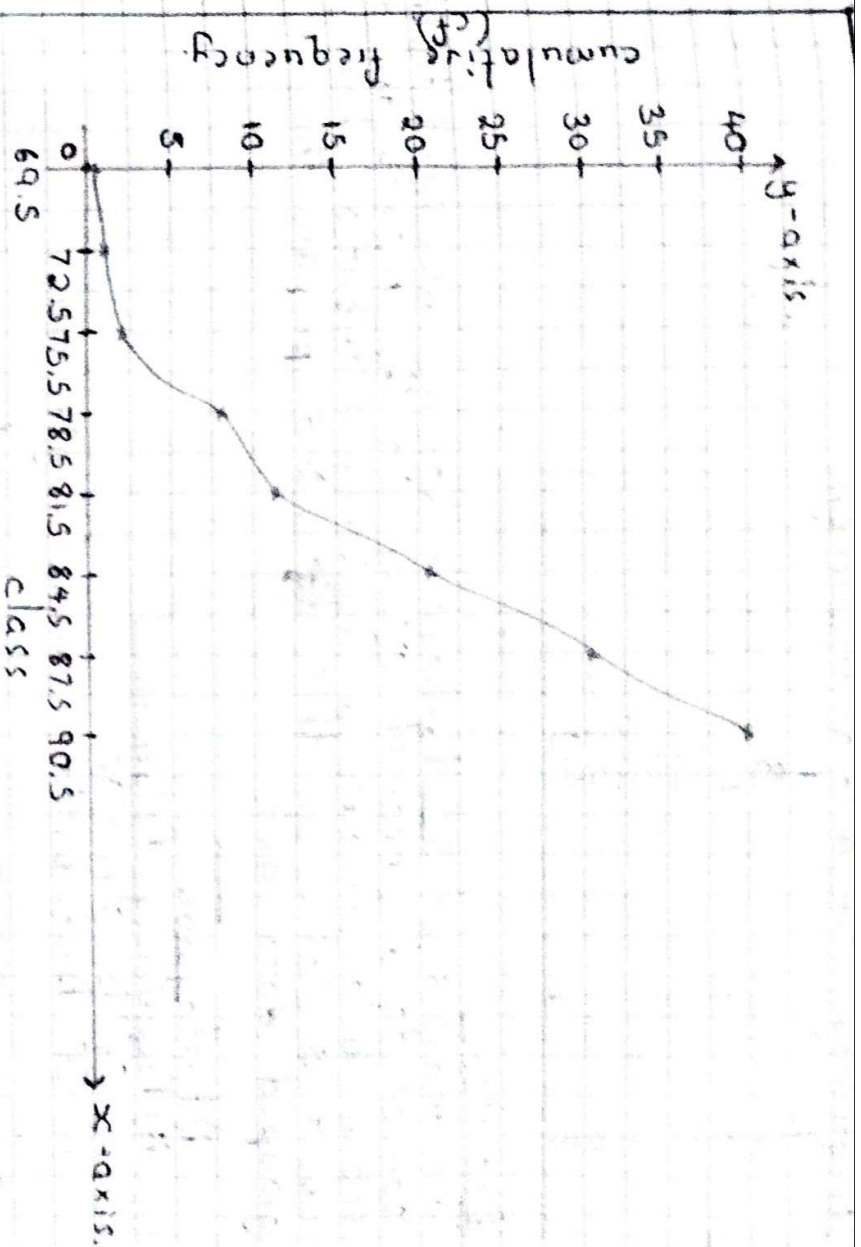


ii) Ogive Curve (Cumulative Frequency Distribution Curve)

An Ogive curve is a graph used to show the cumulative frequency of a dataset.
It is drawn to show by plotting cumulative frequency against class boundaries and then joining the points with a smooth curve.

Exercise

Qn. Use the same data table in the previous Histogram part to draw an ogive curve for it.



iii) Frequency Polygon is a graph produced by plotting frequency against class mid-point values and joining coordinates with a straight line.

Exercise

The frequency polygon is done in the previous page on top of the histogram.

iv) Cumulative Frequency Distribution

It is a table showing cumulative frequency for each value of upper-class boundary. The cumulative frequency of upper class boundary is obtained by adding the frequency of the class to the sum of previous frequencies.

Exercise

AN. Find the C.F. of the above dates.

Frequency	C.F	Total frequency
1		1
2	3	3
7	10	10
12	22	22
9	31	31
6	37	37
3	40	40

Qn. The masses of 50 tyres in kg are measured to the correct to the nearest 0.1 kg and the results are as shown below. Produce a frequency distribution having about 5 classes of this data and prepare the grouped data as :-

- Histogram.
- Frequency Polygon.
- Cf Distribution.
- Cf Distribution curve / Ogive curve.

8.0	8.6	8.2	7.5	8.0	9.1	8.5	7.6	8.2	7.8
8.3	7.1	8.1	8.3	8.7	7.8	8.7	8.5	8.4	8.5
7.7	8.4	7.9	8.8	8.0	8.1	7.8	8.0	7.7	7.5
8.1	7.4	8.8	8.0	8.4	8.5	8.1	7.3	9.0	8.6
7.4	8.2	8.4	7.7	8.3	8.2	7.9	8.5	7.9	8.0

$$\text{Range} = 9.1 - 7.1 = 2$$

$$\frac{7.6 - 7.5}{2} = 0.05$$

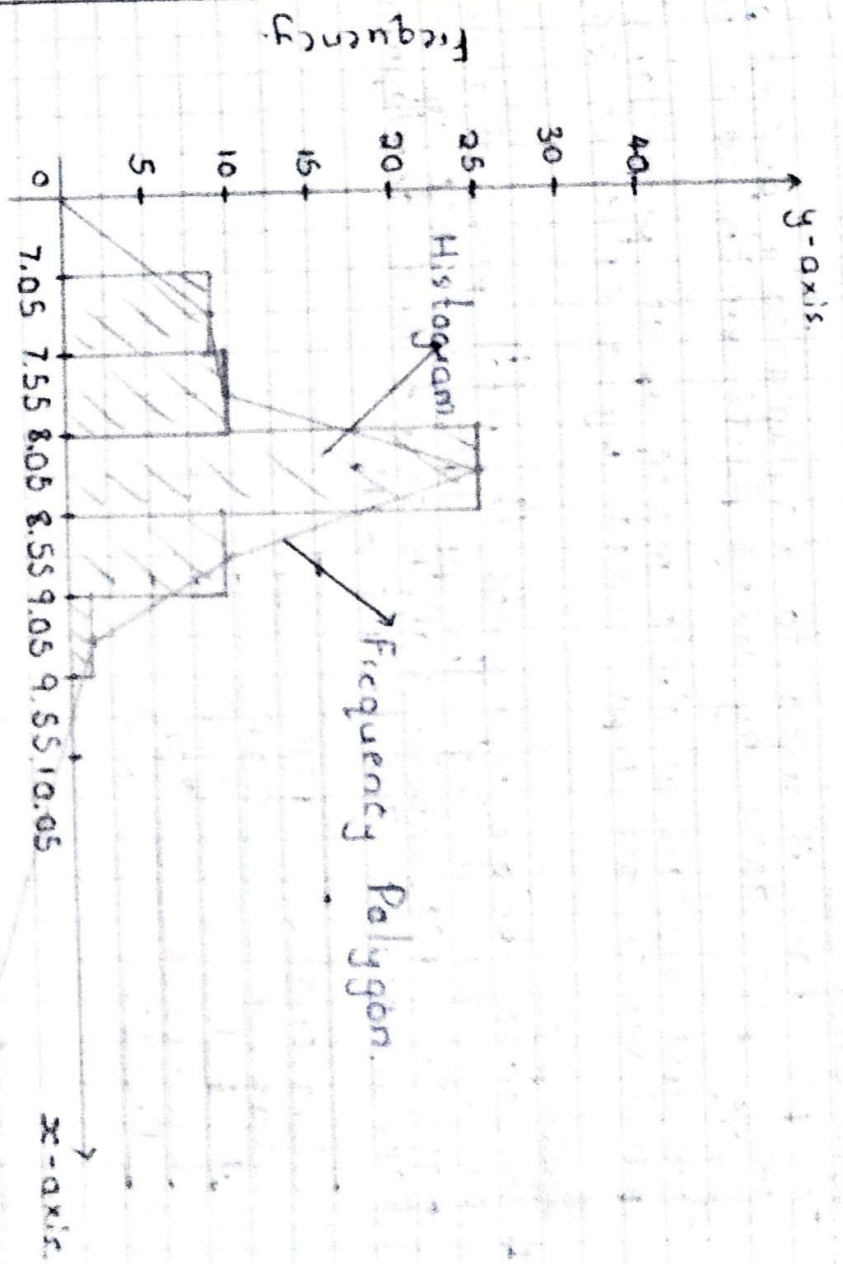
$$\text{Interval} = \frac{2}{5} = 0.4$$

Cumulative

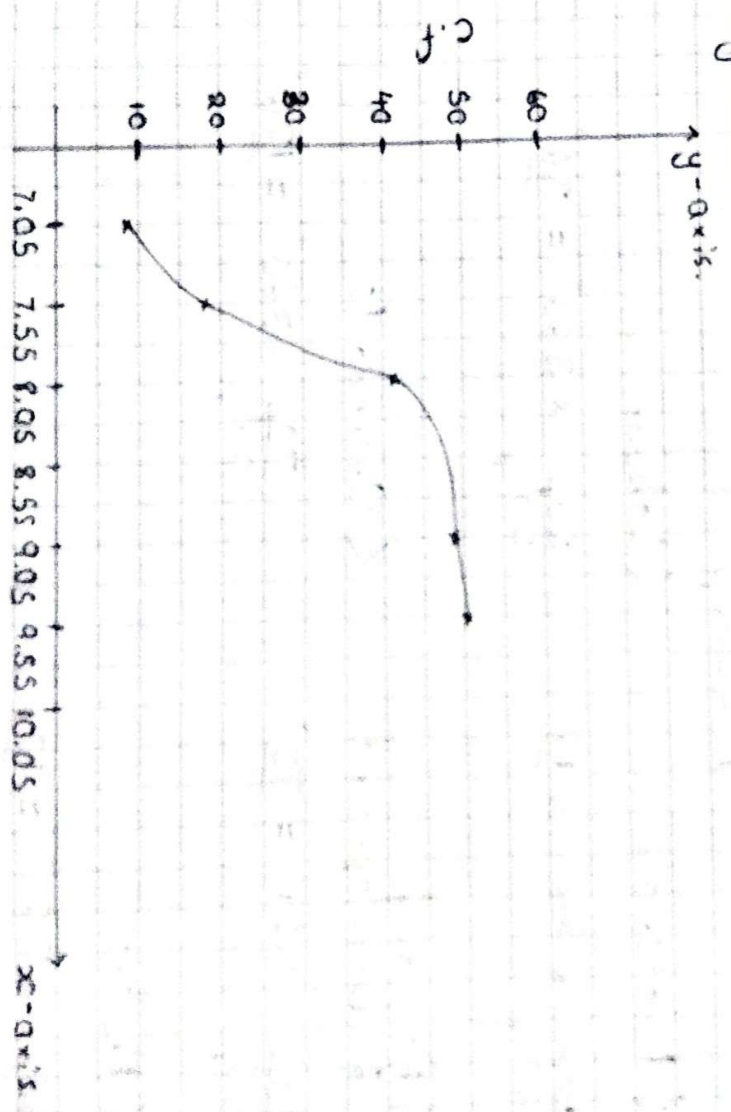
Frequency Distribution

Class	Tally	f	cf	Mid-p.
7.01 - 7.5	### 111	8	8	7.3
7.6 - 8.0	### ###	10	17	7.8
8.1 - 8.5	#####	25	42	8.3
8.6 - 9.0	### 11	7	49	8.8
9.1 - 9.4	1	1	50	9.3

ii) Histogram and Frequency Polygon.



iii) Ogive Curve



Class.

Ungrouped data (forgetting concept)

Pie Chart

A pie chart is used to diagrammatically present and ungrouped data by using parts which total to 360°.

We use sectors of circle made up of proportional parts which are represented by a circle of 360°.

Example The data below represents the retail price of a product costing 2000 shillings. The products is made up of materials 10P, Labour 20P, Research and development 40P, overheads 70P, Profits 60P,

Materials.	10P
Labour	20P
R & D	40P
Overheads	70P
Profits	60P

Research present this diagram on a Pie-Chart.

Total Quantities = 10 + 20 + 40 + 70 + 60

$$T = 200P.$$

Materials

$$\frac{10P}{200P} \times 360^\circ = \underline{\underline{18^\circ}}$$

Labour

$$\frac{20P}{200P} \times 360^\circ = \underline{\underline{36^\circ}}$$

Research & Development.

Overheads

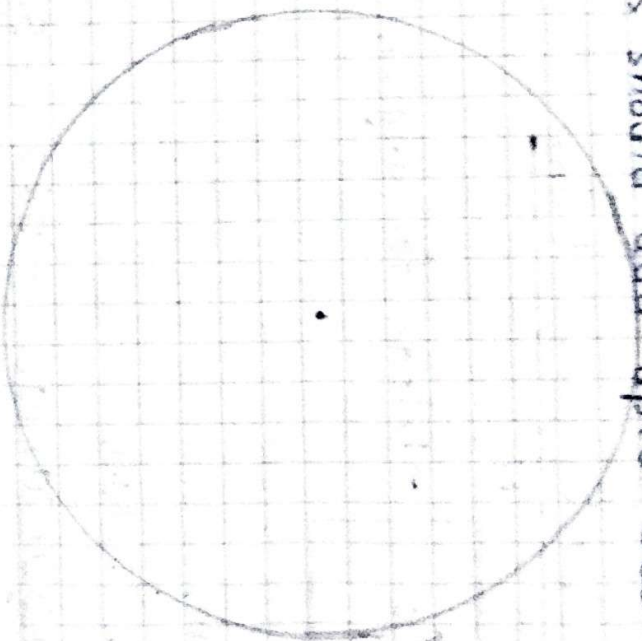
$$\frac{40P}{200P} \times 360^\circ = \underline{\underline{72^\circ}}$$

$$\frac{70P}{200P} \times 360^\circ = \underline{\underline{126^\circ}}$$

Profits.

$$\frac{60P}{200P} \times 360^\circ = \underline{\underline{108^\circ}}$$

Use a protractor to measure the percentage to the spot. totals should add upto 360°



Key must be present.

Key	Symbol
Materials	////
Labour	ccccc
R & D	xxxxx
Overheads	-----
Profits	+++++

Measures of Central Tendency

- A single value which is representing a set of values may be used to give an indication of the general size of members in a set.
- The word average is commonly used to indicate the simple value.
- The statistical term used for average is arithmetic mean / mode.
- Other measures of central tendency are:
 - i) Median
 - ii) Modes

Mean, Mode and Median for Discrete data.

Mean

- Obtained by adding all members of the set, divided by the NO. of members, i.e.

$$\text{Set } \{S_1, S_2, S_3, \dots, S_n\}$$

n = no. of members.

$$\bar{x} = \frac{S_1 + S_2 + S_3 + \dots + S_n}{N}$$

$\Rightarrow$ Mean for ungrouped data.

$$\bar{x} = \frac{\sum x}{N}$$

Exercise

4. Find the mean.

i) 20, 40, 30, 70, 100.

$$\frac{20+40+30+70+100}{5} = \frac{260}{5} = \underline{\underline{52}}$$

ii) 65, 72, 74, 80.

$$\frac{65+72+74+80}{4} = \frac{291}{4} = \underline{\underline{72.75}}$$

iii) 50, 60, 40.

$$\frac{50+60+40}{3} = \frac{150}{3} = \underline{\underline{50}}$$

Median

Median is the middle data in a series and is obtained by arranging the data in magnitude.

• Magnitude means in order.
• If the number of items is an even number you add the two middle items and divide by 2.

Mode

• It is the most occurring value in a set.
• If two values occur with the same frequency the set is called bimodal or that set has 2 modes.

Exercise

4 Find i) the mode of the following data:-

20, 40, 70, 20, 85, 50, 20

$$= \underline{\underline{20}}$$

ii) 50, 70, 80, 50

= 50

iii) 35, 45, 75, 25, 45, 55, 45.

= 45.

2. Find the median.

i) 20, 30, 70, 40, 100 $\Rightarrow$ make order.

20, 30, 40, 70, 100

ii) 25, 30, 35, 40, 47, 59

$\frac{35+40}{2} = \underline{\underline{37.5}}$

Mean, Mode & Median for Grouped Data
 To get the mean of a grouped data we use the formula :-

$$\text{Mean} = \frac{\sum fx}{N/f} \Rightarrow \text{Mean for Grouped Data.}$$

Exercise 1 :-
 Find the mean of the following data.

Marks	NO of Students.	x	fx
0-20	5	10	50
20-40	7	30	210
40-60	11	50	550
40-60	8	70	560
60-80	9	90	810
80-100	40		
$\Rightarrow$			$\Rightarrow 2000$

$\bar{x} = \frac{2000}{40} = \underline{\underline{50}}$

$\bar{x} = \underline{\underline{50}}$

3) Find the mean of the following:

8)

Size of shoes.	1	2	3	4	5
NO of pairs.	15	20	25	30	10

$$\frac{15 + 20 + 25 + 30 + 10}{5} = \underline{\underline{20}}$$

ii)

Temperature.	0-30	30-60	60-90
NO of patient.	5	20	3
f_x	15	45	75
f_x	75	900	225
	$\Rightarrow 28$	$\Rightarrow 135$	$\Rightarrow 1200$

$$\bar{x} = \frac{\sum f_x}{f}$$

$$= \frac{1200}{28}$$

$$\bar{x} = \underline{\underline{42.86}}$$

Median

$$\text{Median} = L + \left(\frac{\frac{N}{2} - C_f}{f} \right) \times h/i$$

where :-

L = Lower boundary of the median class.
 N = Total frequency.
 C_f = Cumulative frequency before the median class.
 f = Frequency of the median class.
 h/i = Class width (Interval size).

Exercise

Find the median:

Marks	No of Stud	cf
0-20	5	5
20-40	7	12
40-60	13	25
60-80	8	33
80-100	2	40
$\Rightarrow 40$		$\Rightarrow$

$$\frac{N}{2} = \frac{40}{2} = 20 \Rightarrow 20 \text{ lies in the cumulative frequency of } 25$$

$$\begin{aligned} L &= 40 \\ N &= 40 \\ \sum f &= 13 \\ f &= 20 \end{aligned}$$

$$L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times i$$

$$40 + \left(\frac{20 - 12}{13} \right) \times 20$$

$$40 + \left(\frac{8}{13} \right) \times 20$$

$$= 52.31$$

Mode

Formula 1

$$\text{Mode} = L + \left(\frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \right) \times i$$

where :

L = Lower class boundary of the modal group.
 f_m = Frequency of the modal group (highest frequency).

f_1 = Frequency of the class before the modal group.

f_2 = Frequency of the class after the modal group.
 i = Class interval (width of the class).

Alternative

Formula 2

$$\text{Mode} = L + \left(\frac{f_2}{f_1 + f_2} \right) \times i$$

Formula 3

$\Rightarrow$ Only when given Median and Mean already.

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

* $\Rightarrow$ Use Formula 1 mostly.

Exercise
1. The Table below shows the distribution of marks obtained by students.

Class.	Frequency.
0-10	40
10-20	6
20-30	10
30-40	14
40-50	8
50-60	3

i) Find the mode of the data.
The highest frequency is 14. (30-40)

$$L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times i$$

~~$$L = 30$$

$$N = 45$$~~

$$= L + \left(\frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \right) \times i$$

$$L = 30$$

$$f_m = 14$$

$$f_1 = 10$$

$$f_2 = 8$$

$$i = 10$$

$$\text{Mode} = 30 + \left(\frac{14 - 10}{(14 - 10) + (14 - 8)} \right) \times 10$$

$$= 30 + \left(\frac{4}{(4) + (6)} \right) \times 10$$

$$= 30 + \left(\frac{4}{10} \right) \times 10$$

$$= 30 + 4$$

$$= \underline{\underline{34}}$$

Quantiles, Deciles, Percentiles.

- When data is divided in 2 equal parts $\rightarrow$ we get the Median.

- When data is divided into 4 equal parts $\rightarrow$ we get Quartiles (Q_1, Q_2, Q_3)

• Each part has size = $\frac{n}{4}$

- When data is divided into 10 equal parts $\rightarrow$ we get Deciles (D_1 to D_9)

• Each part has size = $\frac{n}{10}$

When data is divided into 100 equal parts $\rightarrow$ we get Percentiles (P_1 to P_{99}).

• Each part has size = $\frac{n}{100}$

Where to locate them:

$$Q_1 = \frac{n}{4}$$

$$Q_2 = \frac{2n}{4}$$

$$Q_3 = \frac{3n}{4}$$

so,

$$D_k = \frac{kn}{100} = (k) \text{ parts}$$

$$\boxed{\text{Formula} = \frac{kn}{4}}$$

$$P_k = \frac{kn}{100} = 100 \text{ parts. } k = \text{either } 1, 2, 3$$

where k is either 1, 2, 3 for $Q_1 \Rightarrow k$ is 1
 $Q_2 \Rightarrow k$ is 2
 $Q_3 \Rightarrow k$ is 3.

1. Calculate (Sample data).

Marks	f	cf
30	4	4
50	3	11
60	1	14
90		15
	<u>= 15</u>	

a) Find Q_2 (Median) for ungrouped data.

$$Q_2 = \frac{2n}{4}$$

$$= \frac{2(15)}{4}$$

$$= 7.5 = 8 \text{th item}$$

b) Locate the 8th item.
8th item lies in cf 11 so its 50 marks.
 $\therefore 50, Q_2 = 50.$

Exercise (Grouped data).

1 Calculate:

Marks	f	cf
0-20	5	5
20-40	7	12
40-60	13	25
60-80	8	33
80-100	7	40

Percentile 25
Decile 6
Q₃

i) Find Q₃, for example data getting quartile cf.

$$Q_3 = \frac{3n}{4} = \frac{3(40)}{4} = \underline{\underline{30}}$$

ii) Locate the quartile class.
30 lies in cf frequency 33.
So quartile class is 60-80.

iii) Find Q₃ for grouped data.

$$Q_3 = L + \left(\frac{\frac{3n}{4} - c}{f} \right) \times i$$

L = 60
C = 25
f = 8
i = 20

$$\begin{aligned} Q_3 &= 60 + \left(\frac{3\left(\frac{40}{4}\right) - 25}{8} \right) \times 20 \\ &= 60 + \left(\frac{5}{8} \right) \times 20 \\ &= 60 + 12.5 \\ &= \underline{\underline{72.5}} \end{aligned}$$

iv) Find the Decile D₆.

$$D_6 = \frac{kn}{10} = \frac{6(40)}{10} = \underline{\underline{24}}$$

i) Locate the Decile 6 class.
+ 34 lies in cf 25.
+ Class = 40-60.

ii) Find Decile 6 for grouped data.

$$D_6 = L + \left(\frac{\frac{6n}{10} - c}{f} \right) \times i$$

$$\begin{aligned} L &= 40 \\ c &= 12 \\ f &= 13 \\ i &= 20 \end{aligned}$$

$$\begin{aligned} &= 40 + \left(\frac{6(40) - 12}{13} \right) \times 20 \\ &= 40 + \left(\frac{12}{13} \right) \times 20 \end{aligned}$$

$$= 40 + 18.46$$

$$D_6 = \underline{\underline{58.46}}$$

vii) Find Percentile P_{25} .

$$P_{25} = \frac{25(40)}{100} = \underline{\underline{10}}$$

viii) Locate the class.
+ 10 lies in cf 12
+ Class = 20-40

ix) Find the Percentile P_{25} for grouped data.

$$P_{25} = L + \left(\frac{\frac{25n}{100} - c}{f} \right) \times i$$

$$\begin{aligned} L &= 20 \\ c &= 5 \\ f &= 7 \\ i &= 20 \end{aligned}$$

$$\begin{aligned} &= 20 + \left(\frac{\frac{25 \times 40}{100} - 5}{7} \right) \times 20 \end{aligned}$$

$$= 20 + \left(\frac{5}{7}\right) \times 20$$

$$= 20 + 14.29$$

$$= \underline{\underline{34.29}}$$

Measures of Dispersion:

- i) Range - It is the difference between the highest and lowest value in a dataset.

Example -

77, 75, 30, 90, 25

$$\text{Range} = 90 - 30 = \underline{\underline{60}}$$

- ii) Interquartile Range - Difference b/w the third quartile and first quartile.

$$\text{Interquartile Range} = Q_3 - Q_1$$

- iii) Quartile Deviation - Difference b/w the 3rd quartile and 1st quartile divide by 2.

$$Q.D = \frac{(Q_3 - Q_1)}{2}$$

- iv) Mean Deviation -

Calculate the mean of a set of data $x_1, x_2, x_3, \dots, x_n$.

$$\bar{x} = \frac{\sum x}{n}$$

vi) Variance

It is the average of the squared deviations from the mean.

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

or

For discrete and grouped data.

$$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

vii) Standard Deviation

It is the square root of variance.

$$\sigma = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

or

For grouped data:

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$$

Exercise

1. A researcher collected the following data on the daily earnings (in dollars) of workers.

Class	f	cf	x	fx
10-20	4	4	15	60
20-30	6	10	25	150
30-40	10	20	35	350
40-50	14	34	45	630
50-60	8	42	55	440
60-70	5	47	65	325
70-80	3	50	75	225
	= 50			= 2180

i) Find the Range of the data?

$$\text{Range} = 80 - 10$$

$$= 70$$

ii) Calculate the interquartile range?

$$= Q_3 - Q_1$$

$$\frac{3n}{4} = Q_3 = \frac{3(50)}{4} = 37.5$$

$$= 50 + \left(\frac{37.5 - 34}{8} \right) \times 10 = 54.375$$

$$Q_1 = \frac{n}{4} = \frac{50}{4} = 12.5 \quad \text{class} = 30-40$$

$$= L + \left(\frac{\frac{n}{4} - c}{f} \right) \times i$$

$$= 30 + \left(\frac{12.5 - 10}{10} \right) \times 10$$

$$\text{Interquartile range} = 54.375 - 32.5 =$$

$$\text{Ans} = 21.875$$

iii) Calculate the Quartile Deviation?

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

$$= \frac{54.375 - 32.5}{2}$$

$$= 10.94$$

ii) Calculate the Mean using the assumed mean method.

$$\bar{x} = \frac{\sum fx}{N}$$

$$\bar{x} = \frac{2180}{50}$$

$$= 43.6$$

v) Calculate the variance of the data.

$$\sigma^2 = \frac{\sum f(x_i - \bar{x})^2}{N}$$

Table extension:

x	f	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
15	4	28.6	817.96	3271.84
25	6	-18.6	345.96	2075.76
35	10	-8.6	73.96	739.6
45	14	1.4	1.96	27.44
55	8	11.4	129.96	1039.68
65	5	21.4	457.96	2289.8
75	3	31.4	985.96	2957.88
				<u>12402</u>

$\bar{x} \Rightarrow$ this is the we got previously in the question.
 $x - \bar{x} \Rightarrow$ this is the mid-point subtract mean ($\bar{x}$).
 $f(x - \bar{x})^2 \Rightarrow$ this is obtained by squaring $x - \bar{x}$,
 and then we multiply by f and $(x - \bar{x})^2$.

$$\text{Variance} = \frac{\sum f(x_i - \bar{x})^2}{N}$$

$$\sigma^2 = \frac{12402}{50}$$

$$\sigma^2 = \underline{\underline{248.04}}$$

ii) Find the standard deviation?

$$s.d = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{248.04}$$

$$\sigma = \underline{\underline{15.75}}$$

iii) Comment on the spread of the data?

✓ Is it highly spread or consistent.
✓ Use your standard deviation and IQR to justify.

- + IQR = 12.875 → middle data is moderately spread.
- + Standard Deviation = 15.75 → values are quite spread from the mean.
- The data shows a moderate to high variation, meaning worker's earnings are not very consistent.

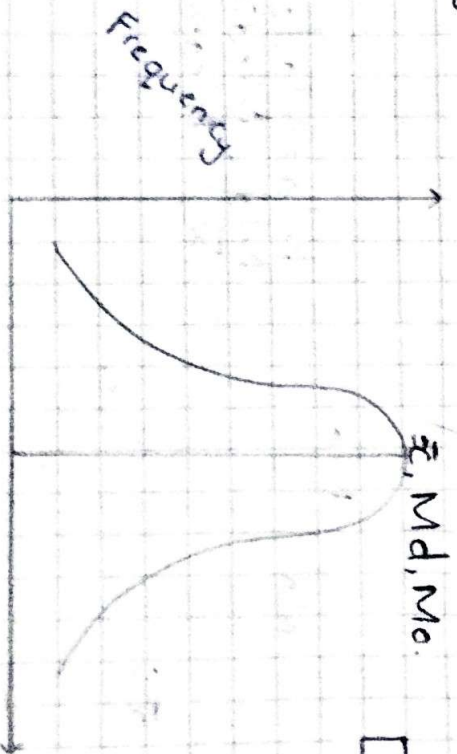
Skewness and Kurtosis

Skewness

- Skewness of a distribution is described as the lack of symmetry.
- In a symmetrical distribution:

$$\text{Mean} = \text{Median} = \text{Mode.}$$

Symmetrical Distribution



$\bar{x}$	$=$	M_d	$=$	M_o
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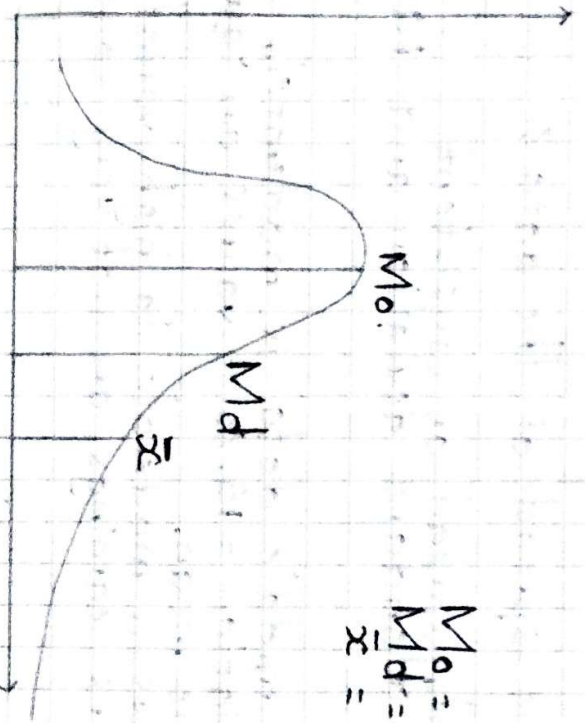
where;
 $\bar{x}$ = Mean (AM)
 M_d = Median
 M_o = Mode.

Symmetrical distribution.

1. Types of Skewness

1. Positive Skew Distribution (Right Skewed)
 - There are extreme observations on the right side of the distribution.
 - The tail is longer on the right.
 - Mean, median and mode are not equal.

$$\boxed{\text{Mean} > \text{Median} > \text{Mode}}$$

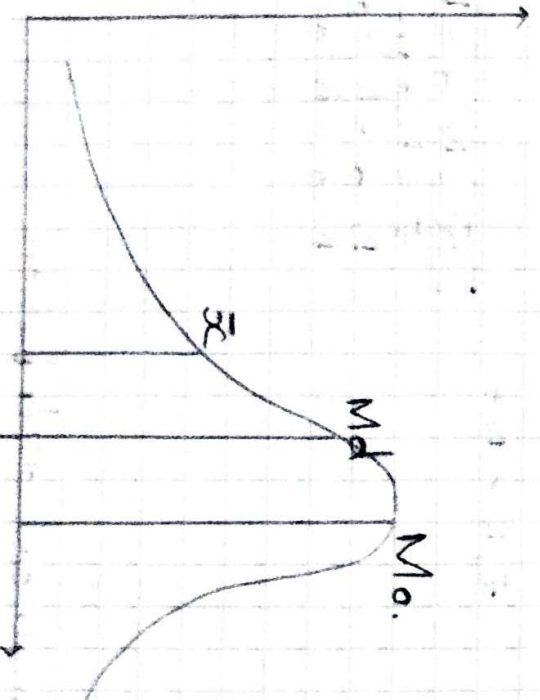


$$\begin{aligned} M_o &= \text{Mode} \\ M_d &= \text{Median} \\ \bar{x} &= \text{Mean} \end{aligned}$$

Positive skew distribution.

2. Negatively Skewed Distribution
 - The tail is longer on the left-hand side.
 - Most data values are connected on the right.

$$\boxed{\text{Mean} < \text{Median} < \text{Mode}}$$



$$\begin{aligned} M_o &= \text{Mode} \\ M_d &= \text{Median} \\ \bar{x} &= \text{Mean} \end{aligned}$$

Measures of Skewness:

Karl Pearson's Measure of Skewness

Karl Pearson came up with 2 formulas.
In formula 1, he considered mean mode and standard deviation.

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

- If the calculation yields:

- i) If $S_k > 0 \Rightarrow$ Positively skewed distribution.
- ii) If $S_k < 0 \Rightarrow$ Negatively skewed distribution.

- The second formula:
Used when mode is not clearly defined.

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Exercise

1. A dataset representing the test scores of students is given below:

Class.	Frequency.
0-10	3
10-20	5
20-30	9
30-40	12
40-50	8
50-60	3

Calculate the mean of the distribution.

Class.	f	x	fx	cf
0-10	3	5	15	3
10-20	5	15	75	8
20-30	9	25	225	17
30-40	12	35	420	29
40-50	8	45	360	37
50-60	3	55	165	40
	<u>40</u>		<u>1260</u>	

median

$$\bar{x} = \frac{\sum fx}{N}$$

$$\bar{x} = \frac{1260}{40}$$

$$\bar{x} = \underline{\underline{31.5}}$$

Find the variance.
Extend the table.

x	f	x - $\bar{x}$	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
5	3	-26.5	702.25	2106.75
15	5	-16.5	272.25	1361.25
25	9	-6.5	42.25	380.25
35	12	3.5	12.25	147
45	8	13.5	182.25	1458
55	3	23.5	552.25	1656.75
	<u>40</u>			<u>7109</u>

$$\text{Variance} = \frac{\sum f(x - \bar{x})^2}{N}$$

$$\sigma^2 = \frac{7109}{40}$$

$$\sigma^2 = \underline{\underline{177.3}}$$

iii) Find the standard deviation,

$$\sigma = \sqrt{177.73}$$

$$\sigma = \underline{\underline{13.33}}$$

iv) Determine the median of the distribution,

$$\frac{N}{2} = \frac{40}{2} = \underline{\underline{20}}$$

Median class = 30-40

$$= L + \left(\frac{\frac{N}{2} - c}{f} \right) \times i$$

$$= 30 + \left(\frac{20 - 17}{12} \right) \times 10$$

$$= 30 + 2.5$$

$$= \underline{\underline{32.5}}$$

v) Using Karl Pearson's formula; Calculate the coefficient of skewness.

$$S_k = 3 \left(\frac{\text{Mean} - \text{Median}}{\text{Standard deviation}} \right)$$

$$S_k = 3 \left(\frac{31.5 - 32.5}{13.33} \right)$$

$$S_k = \frac{-3}{13.33}$$

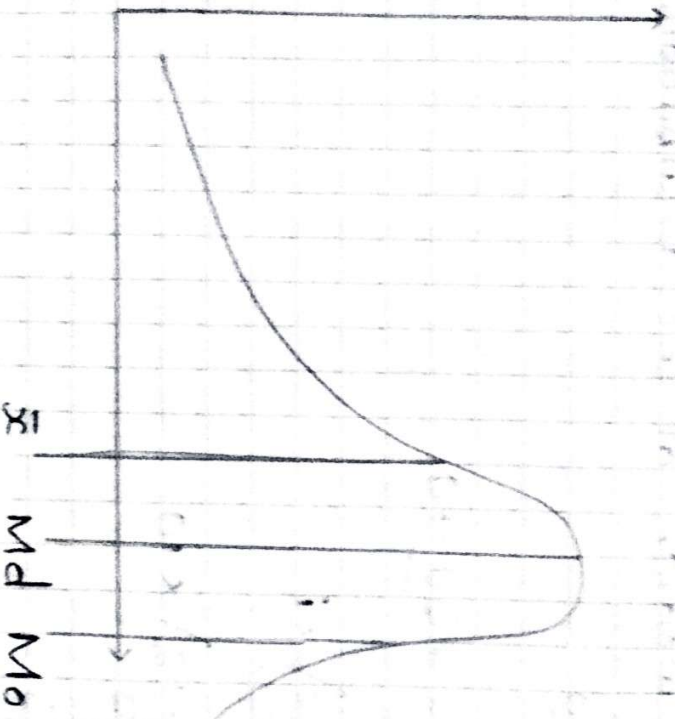
$S_k = -0.223 \Rightarrow$ Not greater than 0.

vi) Whether the distribution is positively skewed, negatively skewed or asymmetrical.

⇒ $Sk = -0.23$ (It is less than 0) therefore having a negative skewed distribution.

ii) The relationship b/w the Mean & Median.
Mean < Median.

iii) Draw the representation in a distribution.



Mean < Median.

Bowley's Measure of Skewness.

$$sq = \frac{Q_3 - 2Md + Q_1}{Q_3 - Q_1}$$

→ This formula measures skewness (asymmetry) using quartiles.
where is

Q_1 = lower quartile (25%)

Md = median (50%)

Q_3 = upper quartile (75%)

$S_q = 0 \rightarrow$ symmetry distributing.
 $S_q > 0 \rightarrow$ positively skewed (tail to the right).
 $S_q < 0 \rightarrow$ negatively skewed (tail to the left).

This method is useful when data has extreme values (outliers).

Outliers $\Rightarrow$ is a value in a dataset that is very different from the rest of the values. i.e.

2, 3, 4, 5, 6, 100

100 is an outlier.

Types of Outliers

i) High outliers $\rightarrow$ much bigger than others.

ii) Low outliers $\rightarrow$ much lower than others.

Bowley's skewness is useful because it is less affected by outliers.
Mean-based methods are more affected by outliers.

Kelly's Measure of Skewness

$$S_p = \frac{P_{90} - 2P_{50} + P_{10}}{P_{90} - P_{10}}$$

This uses percentiles instead of quartiles.
where;

P_{10} = 10th percentile.

P_{50} = median

P_{90} = 90th percentile.

Why use it?

- It gives a more detailed measure of skewness because it uses more spread-out points.

Moments

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r$$

$$\mu_1 = 0$$

$$\mu_2 = \text{variance}$$

$$\mu_3 = \text{skewness}$$

What is a "moment"?

- A moment tells you how data is distributed around the mean.
- where;

x_i = data value.

$\bar{x}$ = mean

f_i = frequency

r = order of payment.

Different Moments

i) $r=0$

$$\mu_0 = \frac{1}{N} \sum f_i (x_i - \bar{x})^0$$

- Since anything to power 0 = 1, so, $\mu_0 = 1$.

ii) $r=1$

$$\mu_1 = \frac{1}{N} \sum f_i (x_i - \bar{x})$$

- This equals 0.
- Why? Because deviations from the mean always cancel out.

iii) $r=2$

$$\mu = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$

• This is variance.

- Measures spread of data;
- Always positive.

$$r = 3$$

$$U_3 = \frac{1}{N} \sum f_i (x_i - \bar{x})^3$$

• Used to measure skewness.

• Positive $\rightarrow$ right skew.

• Negative $\rightarrow$ left skew.

Kurtosis

• It is another measure of the shape of a distribution.

• The difference between skewness and Kurtosis is that skewness measures the lack of symmetry of a frequency curve of a distribution while kurtosis measure the relative peakedness of a distribution.

Skewness $\rightarrow$ symmetry.

Kurtosis $\rightarrow$ how sharp or flat the peak is.

Types of Kurtosis



Types of Kurtosis

1. Leptokurtic

- + Very sharp peak.
- + High kurtosis.
- + More extreme values.

* Tall & thin graph.

2. Mesokurtic

- + Normal shape (like a bell curve), i.e. normal distribution.

3. Platykurtic

- + Flat peak,
- + Low kurtosis.

* Wide and short graph.

Exercise

1. The following data represents a distribution.

x	2	4	6	8	10
f	3	7	10	7	3

Tasks

1. Find the mean.
2. Calculate the 2nd and 3rd moments about the mean.
3. Determine the skewness using moments.
4. Comment on the kurtosis (shape).
5. Estimate Bowley's co-efficient of skewness.

Answer

x	f	fx
2	3	6
4	7	28
6	10	60
8	7	56
10	3	30
	<u>= 30</u>	<u>= 180</u>

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$= \frac{180}{30}$$

$$\bar{x} = \underline{\underline{6}}$$

2. 2nd and 3rd moments about mean.

$$\mu_r = \frac{1}{N} \sum f(x - \bar{x})^r$$

2nd

x	f	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
2	3	16	48
4	7	4	28
6	10	0	0
8	7	4	28
10	3	16	48
	<u>30</u>		<u>152</u>

$$\sum f(x - \bar{x})^2 = \underline{\underline{152}}$$

$$\mu_2 = \frac{152}{30}$$

$$\mu_2 = \underline{\underline{5.07}}$$

3rd

x	f	$(x - \bar{x})^3$	$f(x - \bar{x})^3$
2	3	-64	-192
4	7	-8	-56
6	10	0	0
8	7	8	56
10	3	64	192
	<u>30</u>		<u>0</u>

$$\sum f(x - \bar{x})^3 = 0$$

$$\mu = \frac{0}{30}$$

$$\mu = \underline{\underline{0}}$$

3. Skewness using moments.

$$\text{Skewness} = \frac{\mu_3}{\mu_2^{3/2}}$$

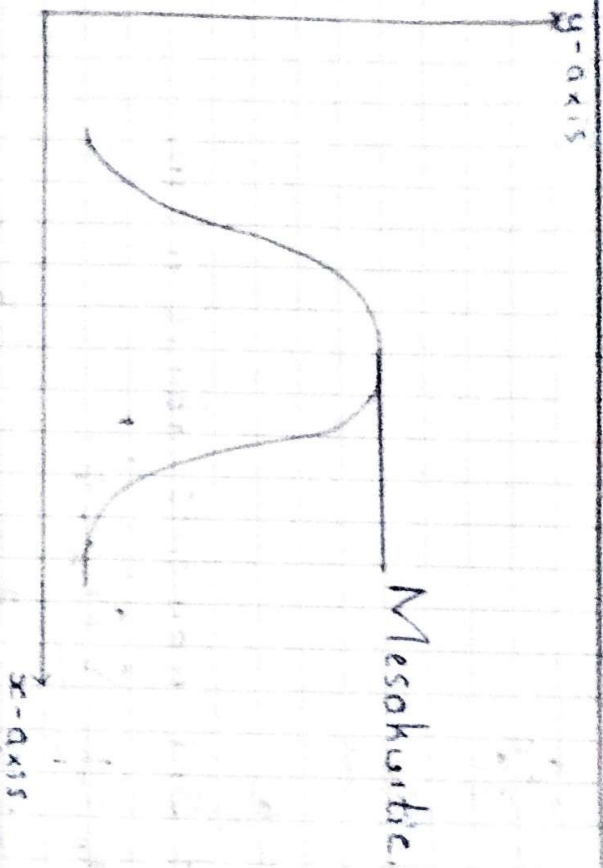
$$= \frac{0}{5.07^{3/2}}$$

$= \underline{\underline{0}} \rightarrow$ Distribution is symmetrical.

4. Comment on the kurtosis (shape).

- Since the data is perfectly symmetric and evenly spread:

- It is approximately Mesokurtic (normal shape).



5. Bowley's co-efficient of skewness,

$$S_q = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

$$Q_1 = \frac{N}{4} = \frac{30}{4} = 7.5 \Rightarrow 4$$

$$Q_2 = \frac{2N}{4} = \frac{30(2)}{4} = 15 \Rightarrow 6$$

$$Q_3 = \frac{3N}{4} = \frac{3(30)}{4} = 22.5 \Rightarrow 8$$

$$S_q = \frac{8 - 2(6) + 4}{8 - 4}$$

$$S_q = \frac{8 - 12 + 4}{4} = \frac{0}{4}$$

$$S_q = \underline{\underline{0}}$$

Regression Analysis

- It is a technique used to show relationship b/w 2 variables.
- We have different methods of getting a regression line:

- i) Inspection Method.
- ii) Least square Method.
- iii) Semi-average Method.

Inspection Method

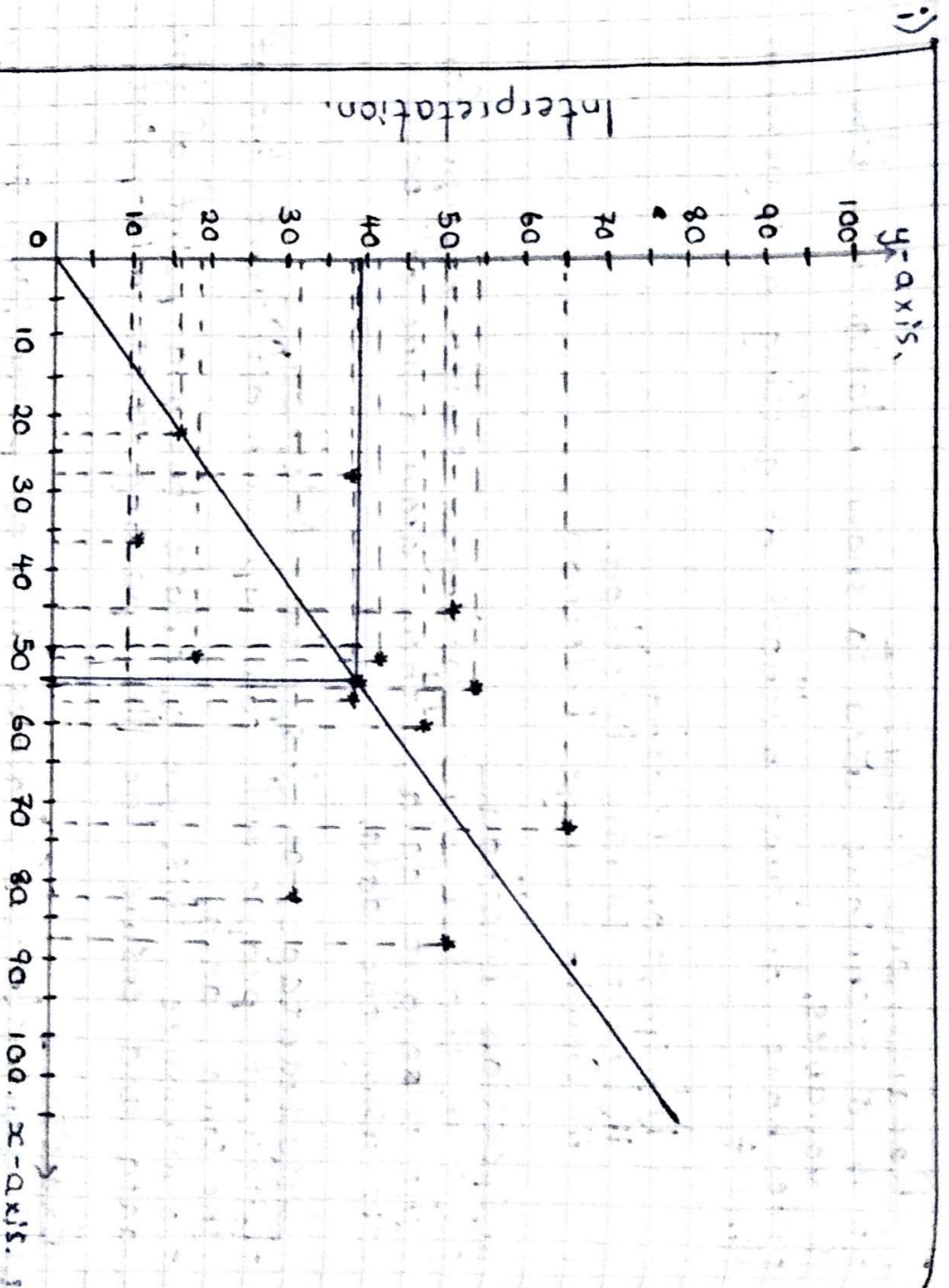
- This method involves plotting a scatter diagram for the relevant data, then draw a line of best fit using the data.
- Disadvantage of Inspection Method
- Different people will come up with different line of the same data.
- To minimize this, we calculate the mid-point of the data and ensure the line passing that mid-point.

Exercise:

1. 12 administrative trainees in a company took an aptitude test in 2 parts and decided to test ability to do appropriate calculations and the other decided to test skills in interpreting results.

Calculation Score	23	56	74	29	82	45	36	51	60	55
Interpretation	16	38	65	39	32	51	11	19	47	54
	52	88								
	43	50								

- i) Draw a scatter-diagram?
- ii) Draw a mid-point then draw a regression line.
- iii) Find the calculation score when the interpretation is 10.
- iv) Find the interpretation when the calculation score is 50.



∴ Calculation Score

ii) To get the mid-point, we add values of x axis, (Calculation Score) divide by the f to find the mean of x for value $\bar{x}$, the also do the same to y find mean of y axis (Interpretation) to get value

∴ draw a line of regression on the values gotten.

$$\bar{x} = \frac{23 + 56 + 74 + 29 + 82 + 45 + 36 + 51 + 60 + 55 + 52 + 88}{12}$$

$$= \underline{\underline{54}}$$

$$\bar{y} = \frac{16 + 38 + 65 + 39 + 32 + 51 + 11 + 19 + 47 + 54 + 43 + 50}{12}$$

$$= \underline{\underline{39}}$$

(54, 39) ⇒ plot a regressor line plot.

- iii) Find the calculation score when the interpretation is 10. $\Rightarrow 54$.
- iv) Find the interpretation when the calculation score is 50. $\Rightarrow 39$.

Semi-Average Method (Trend analysis)

The Semi-Average Method is a technique used to determine the trend line in a time series by dividing the data into 2 equal parts and finding the average of each part.

Purpose - To identify the general direction (trend) of data over time.

Steps involved

1. Divide the data into 2 equal parts $\rightarrow$ split into 2 equal groups. If odd $\rightarrow$ leave out the middle value.
2. $x \rightarrow$ to be ascending then $y \rightarrow$ follows x but don't change arrangement.
3. Find the average of each group; Calculate:
 - i) First half average $\rightarrow \bar{Y}_1$
 - ii) Second half average $\rightarrow \bar{Y}_2$

4. Determine the mid-points of time.
5. Plot the 2 points.

$$\text{Plot} \Rightarrow (X_1, \bar{Y}_1)$$

$$(X_2, \bar{Y}_2)$$

6. Draw the line of trend.
 - Join the 2 points with a straight line.
 - This gives a trend line.

Formula of the Trend line

$$\boxed{Y = a + bx}$$

where;

a = intercept
 b = slope (rate of change).

Exercise.

Calculation score	23	56	74	29	82	45	36	51	60	55
Interpretation	16	39	65	39	32	51	11	19	47	54
	52	88								
	43	50								

Group A	
x	y
23	16
29	39
36	11
45	51
51	19
52	43
$\Sigma = 236$	$\Sigma = 179$

Group B	
x	y
53	54
56	38
60	47
74	65
82	32
88	50
$\Sigma = 415$	$\Sigma = 286$

$$\bar{x} = \frac{236}{6} = \underline{\underline{39}}$$

$$\bar{x} = \frac{415}{6} = \underline{\underline{69}}$$

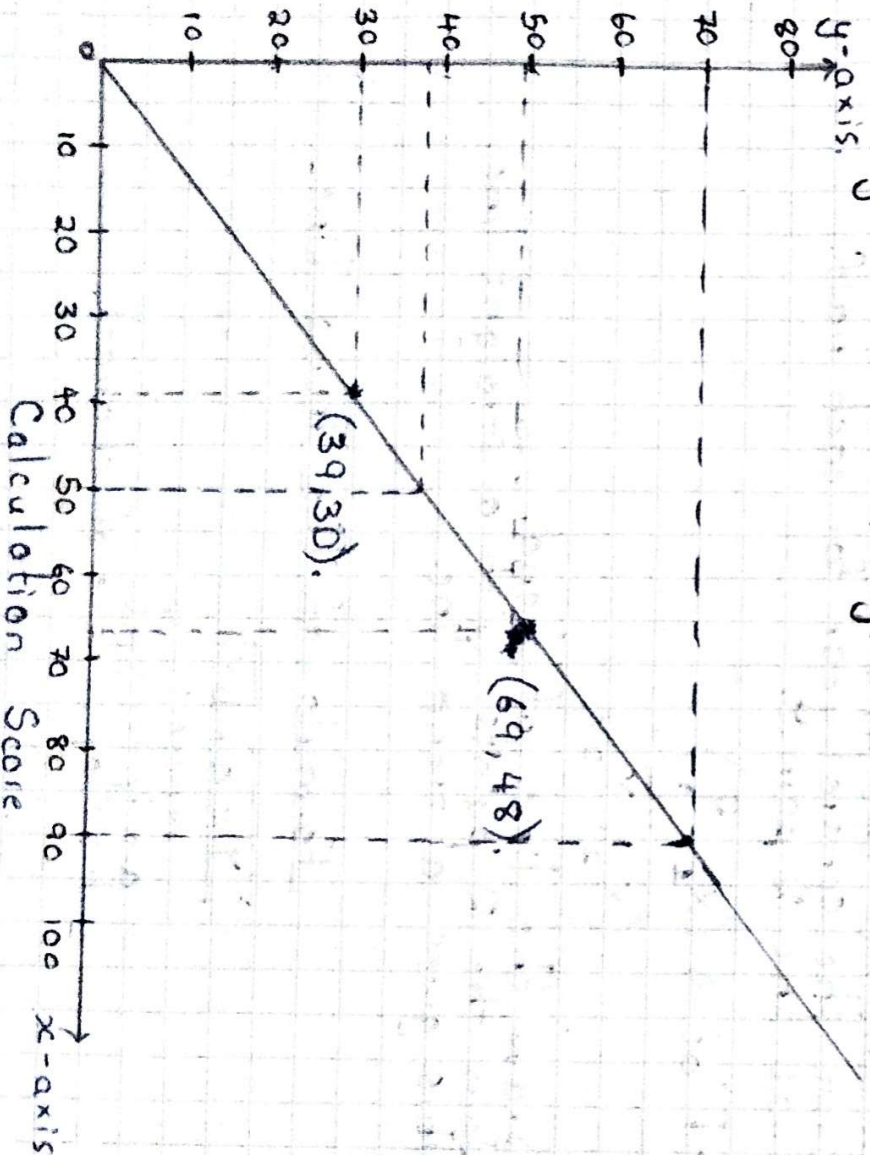
$$\bar{y} = \frac{179}{6} = \underline{\underline{30}}$$

$$\bar{y} = \frac{286}{6} = \underline{\underline{48}}$$

$= (39, 30)$ and $(69, 48)$

x
y
y-axis

Interpretation.



- i) Find the interpretation ~~more~~ when calculation score is 50? $\Rightarrow 38$.
- ii) Find the $\approx$ calculation score when Interpretation is 70? $\Rightarrow 90$.

Advantages.

1. Simple and easy to use.
2. Quick method for finding trends.
3. Requires minimal calculation.

Disadvantages.

1. Not very accurate.
2. Ignores fluctuations in groups.
3. Requires ~~minimal~~ Depends heavily on how data is divided.

Least Square Method.

first use $\sum x$ & $\sum y$ to get $\sum xy$ & $\sum x^2$

$$b = \frac{n \cdot \sum xy - \sum x \cdot \sum y}{n \cdot \sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y - b \sum x}{n}$$

where;

a = Intercept (value of y when $x = 0$) with respect to x

b = Slope (rate of change of y with respect to x)

X = Independent variable (Age).

Y = Dependent variable (Cost).

n = Number of observations.

$\sum x$ = Sum of all x values.

$\sum y$ = Sum of all y values.

$\sum xy$ = Sum of products of x and y .

$\sum x^2$ = Sum of squares of x .

Exercise

1. Given Data:

Age (x)	5	10	15	20	30	30	50	50	60
Cost (y)	100	200	250	300	310	335	300	380	395

x	y	xy	x ²	y ²
5	118	550	25	12100
10	200	2000	100	40000
15	250	3750	225	62500
20	300	6000	400	90000
30	310	9300	900	96100
30	335	10050	900	112225
50	300	15000	2500	90000
50	380	19000	2500	144400
60	395	23700	3600	156025
Σ = 300	Σ = 2970	Σ = 97650	Σ = 12050	

$$b = \frac{n \cdot \sum xy - \sum x \cdot \sum y}{n \cdot \sum x^2 - (\sum x)^2}$$

$$\sum xy = 97650$$

$$\sum x = 300$$

$$\sum y = 2970$$

$$\sum x^2 = 12050$$

$$n = 10$$

$$b = \frac{10(97650) - (300) \cdot (2970)}{10(12050) - (300)^2}$$

$$b = \frac{976500 - 891000}{120500 - 90000}$$

$$b = \frac{85500}{30500}$$

$$b = \underline{\underline{2.8}}$$

$$a = \frac{\sum y - b \sum x}{n}$$

$$a = \frac{2970 - 2.8(300)}{10}$$

$$a = \frac{2130}{10}$$

$$a = 213$$

$$\text{Final equation} = Y = \underline{\underline{213}} + 2.8X$$

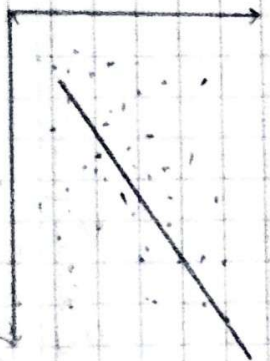
$$Y = \underline{\underline{213}} + 2.8X$$

Correlation

- Correlation is a statistical measure that shows the degree and direction of relationship b/w 2 variables.
- Correlation is a number b/w -1 and 1, that the increase in the value of 1 variable tend to be associated with increase of the other.
- If r is negative (-) tend mean that increase in the value of 1 variable tend to be associated with decrease variable of the other variable.

Types of Correlation

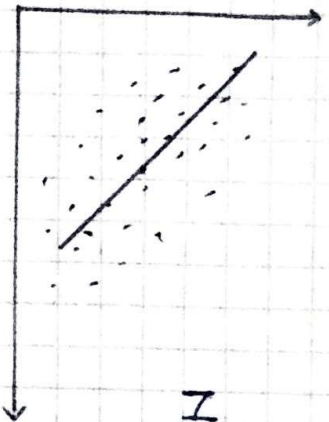
1. Positive Correlation
When one variable increases, the other also increases.



Positive
Correlation.

2. Negative Correlation

- When one variable increases, the other decreases.



Negative Correlation.

3. Zero Correlation

- No relationship btwn variables.



Zero Correlation.

Coefficient of Correlation (r)

- Measures the strength and direction of relationship.

- Value lies btwn:

$$-1 \leq r \leq +1$$

- $+1 \rightarrow$ perfect positive correlation.
- $0 \rightarrow$ no correlation.
- $-1 \rightarrow$ perfect negative correlation.

Karl Pearson's Coefficient

$$r = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

where;

r = correlation coefficient.

x, y = variables.

n = number of observations.

$\sum x$ = sum of x values.

$\sum y$ = sum of y values.

$\sum xy$ = sum of products.

$\sum x^2, \sum y^2$ = sum of squares.

Steps to Calc. correlation.

1. Prepare a table:

x	y	xy	x^2	y^2
-----	-----	------	-------	-------

2. Find

$$\sum x, \sum y, \sum xy, \sum x^2, \sum y^2.$$

3. Substitute in the formula.

4. Get r .

Properties of Correlation.

$$-1 \leq r \leq +1$$

- It is unit free.

- Measures only linear relationship.

- Symmetrical

$$r_{xy} = r_{yx}$$

Advantages.

- Simple to understand.

- Useful in prediction.

- Shows both strength and direction.

Disadvantages.

- Only measures linear relationships.

- Does not imply causation.

- Can be affected by extreme values (outliers).

Spearman's Rank Correlation.

- Rank correlation is a statistical measure that determines the degree and direction of relationship between two variables based on their ranks rather than their actual values.

It shows how well the order (ranking) of one variable matches the order of another variable. It is an alternative method of measuring correlation using the ranks of the values instead of actual values.

5

- [illegible]

- ⇒ The ranking of x_i values is done independently of the ranking of y values.
- ⇒ The ranking is normally done in ascending order.
- ⇒ Find the difference b/w ranks and square the difference.

$$p = (r_x - r_y)$$

$\Rightarrow$ Least square.

$$r' = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

is a constant.

⇒ Rank correlation coefficient.

Exercise

a. Given Data below, calculate the Karl Pearson's

5	10
4	8
3	6
2	4
1	2
0	0

x	y	x ²	y ²	xy
2	4	4	16	8
4	6	16	36	24
6	8	36	64	48
8	10	64	100	80
10	12	100	144	120
12	14	144	196	168
14	16	196	256	224
16	18	256	324	288
18	20	324	400	360
20	22	400	484	440
22	24	484	576	528
24	26	576	676	624
26	28	676	784	728
28	30	784	900	840
30	32	900	1024	960
32	34	1024	1156	1088
34	36	1156	1296	1224
36	38	1296	1444	1368
38	40	1444	1600	1520
40	42	1600	1764	1680
42	44	1764	1936	1848
44	46	1936	2116	2024
46	48	2116	2304	2208
48	50	2304	2500	2400
50	52	2500	2704	2600
52	54	2704	2916	2808
54	56	2916	3136	3024
56	58	3136	3364	3248
58	60	3364	3600	3480
60	62	3600	3844	3720
62	64	3844	4096	3968
64	66	4096	4356	4224
66	68	4356	4624	4488
68	70	4624	4900	4760
70	72	4900	5184	5040
72	74	5184	5476	5328
74	76	5476	5776	5624
76	78	5776	6084	5928
78	80	6084	6400	6240
80	82	6400	6724	6560
82	84	6724	7056	6888
84	86	7056	7400	7224
86	88	7400	7764	7568
88	90	7764	8136	7920
90	92	8136	8516	8280
92	94	8516	8916	8648
94	96	8916	9336	9024
96	98	9336	9776	9408
98	100	9776	10240	9800
100	102	10240	10724	10200
102	104	10724	11224	10608
104	106	11224	11744	11024
106	108	11744	12284	11448
108	110	12284	12844	11880
110	112	12844	13424	12320
112	114	13424	14024	12768
114	116	14024	14644	13224
116	118	14644	15284	13696
118	120	15284	15944	14176
120	122	15944	16624	14664
122	124	16624	17324	15160
124	126	17324	18044	15664
126	128	18044	18784	16176
128	130	18784	19544	16696
130	132	19544	20324	17224
132	134	20324	21124	17760
134	136	21124	21944	18304
136	138	21944	22784	18856
138	140	22784	23644	19416
140	142	23644	24524	20000
142	144	24524	25424	20592
144	146	25424	26344	21192
146	148	26344	27284	21800
148	150	27284	28244	22416
150	152	28244	29224	23040
152	154	29224	30224	23672
154	156	30224	31244	24312
156	158	31244	32284	24960
158	160	32284	33344	25616
160	162	33344	34424	26280
162	164	34424	35524	26952
164	166	35524	36644	27632
166	168	36644	37784	28320
168	170	37784	38944	29016
170	172	38944	40124	29720
172	174	40124	41324	30432
174	176	41324	42544	31152
176	178	42544	43784	31880
178	180	43784	45044	32616
180	182	45044	46324	33360
182	184	46324	47624	34112
184	186	47624	48944	34872
186	188	48944	50284	35640
188	190	50284	51644	36416
190	192	51644	53024	37200
192	194	53024	54424	37992
194	196	54424	55844	38792
196	198	55844	57284	39600
198	200	57284	58744	40416
200	202	58744	60224	41240
202	204	60224	61724	42072
204	206	61724	63244	42912
206	208	63244	64784	43760
208	210	64784	66344	44616
210	212	66344	67924	45480
212	214	67924	69524	46352
214	216	69524	71144	47232
216	218	71144	72784	48120
218	220	72784	74444	49016
220	222	74444	76124	49920
222	224	76124	77824	50832
224	226	77824	79544	51752
226	228	79544	81284	52680
228	230	81284	83044	53616
230	232	83044	84824	54560
232	234	84824	86624	55512
234	236	86624	88444	56472
236	238	88444	90284	57440
238	240	90284	92144	58416
240	242	92144	94024	59400
242	244	94024	95924	60392
244	246	95924	97844	61392
246	248	97844	99784	62400
248	250	99784	101744	63416
250	252	101744	103724	64440
252	254	103724	105724	65472
254	256	105724	107744	66512
256	258	107744	109784	67560
258	260	109784	111844	68616
260	262	111844	113924	69680
262	264	113924	116024	70752
264	266	116024	118144	71832
266	268	118144	120284	72920
268	270	120284	122444	74016
270	272	122444	124624	75120
272	274	124624	126824	76232
274	276	126824	129044	77352
276	278	129044	131284	78480
278	280	131284	133544	79616
280	282	133544	135824	80760
282	284	135824	138124	81912
284	286	138124	140444	83072
286	288	140444	142784	84240
288	290	142784	145144	85416
290	292	145144	147524	86600
292	294	147524	149924	87792
294	296	149924	152344	88992
296	298	152344	154784	90200
298	300	154784	157244	91416
300	302	157244	159724	92640
302	304	159724	162224	93872
304	306	162224	164744	95112
306	308	164744	167284	96360
308	310	167284	169844	97616
310	312	169844	172424	98880
312	314	172424	175024	100152
314	316	175024	177644	101432
316	318	177644	180284	102720
318	320	180284	182944	104016
320	322	182944	185624	105320
322	324	185624	188324	106632
324	326	188324	191044	107952
326	328	191044	193784	109280
328	330	193784	196544	110616
330	332	196544	199324	111960
332	334	199324	202124	113312
334	336	202124	204944	114672
336	338	204944	207784	116040
338	340	207784	210644	117416
340	342	210644	213524	118800
342	344	213524	216424	120192
344	346	216424	219344	121592
346	348	219344	222284	123000
348	350	222284	225244	124416
350	352	225244	228224	125840
352	354	228224	231224	127272
354	356	231224	234244	128712
356	358	234244	237284	130160
358	360	237284	240344	131616
360	362	240344	243424	133080
362	364	243424	246524	134552
364	366	246524	249644	136032
366	368	249644	252784	137520
368	370	252784	255944	139016
370	372	255944	259124	140520
372	374	259124	262324	142032
374	376	262324	265544	143552
376	378	265544	268784	145080
378	380	268784	272044	146616
380	382	272044	275324	148160
382	384	275324	278624	149712
384	386	278624	281944	151272
386	388	281944	285284	152840
388	390	285284	288644	154416
390	392	288644	292024	156000
392	394	292024	295424	157592
394	396	295424	298844	159192
396	398	298844	302284	160800
398	400	302284	305744	162416
400	402	305744	309224	164040
402	404	309224	312724	165672
404	406	312724	316244	167312
406	408	316244	319784	168960
408	410	319784	323344	170616
410	412	323344	326924	172280
412	414	326924	330524	173952
414	416	330524	334144	175632
416	418	334144	337784	177320
418	420	337784	341444	179016
420	422	341444	345124	180720
422	424	345124	348824	182432
424	426	348824	352544	184152
426	428	352544	356284	185880
428	430	356284	360044	187616
430	432	360044	363824	189360
432	434	363824	367624	191112
434	436	367624	371444	192872
436	438	371444	375284	194640
438	440	375284	379144	196416
440	442	379144	383024	198200
442	444	383024	386924	200000
444	446	386924	390844	201808
446	448	390844	394784	203624
448	450	394784	398744	205448
450	452	398744	402724	207280
452	454	402724	406724	209120
454	456	406724	410744	210968
456	458	410744	414784	212824
458	460	414784	418844	214688
460	462	418844	422924	216560
462	464	422924	427024	218440
464	466	427024	431144	220328
466	468	431144	435284	222224
468	470	435284	439444	224128
470	472	439444	443624	226040
472	474	443624	447824	227960
474	476	447824	452044	

2. Given data below, calculate the Rank Correlation (Spearman's).

x	10	20	30	40	50
y	8	6	9	5	7

Rank x	Rank x	y	Rank y
10	1	8	4
20	2	6	2
30	3	9	5
40	4	5	1
50	5	7	3

Find d^2

$$d^2 = (\underline{r_x - r_y})^2$$

Rank x	Rank y	d	d^2
1	4	-3	9
2	2	0	0
3	5	-2	4
4	1	3	9
5	3	2	4
$= 26$			

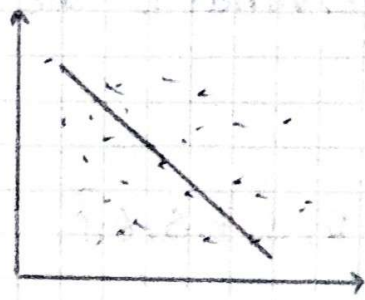
$$d^2 = \underline{26}$$

$$r = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$r = \frac{1 - 6(26)}{5(25 - 1)}$$

$$r = 1 - 1.3 = \underline{-0.3}$$

$r = -0.3 \rightarrow$ weak negative correlation.



negative correlation.

Probability.

- It is the likelihood of a event to occur.
- It is a number between 0 and 1.

$$P(A) = \frac{\text{NO of ways event A can occur.}}{\text{The Total number of possible outcomes.}}$$

- Define the following terms :

- i) Outcome - an outcome, is a single possible result of a random experiment.
- ii) Event - is a set of one or more outcomes of an experiment.
- iii) Sample space - is the set of all possible outcomes of a random experiment.

Example.

1. If you toss a coin once, what is the probability of getting A.

Sample space = {Head, Tail}

Total NO of possible outcomes = 2

$$P(H) = \frac{\text{NO of ways of landing at H}}{\text{Total NO of possible outcomes}}$$

$$\Rightarrow P(H) = \frac{1}{2}$$

$$\Rightarrow P(T) = \frac{1}{2}$$

2. If you roll a 6-sided dice. What is the probability of getting :-

i) An even number.

Sample space = $\{1, 2, 3, 4, 5, 6\}$

$$P(E) = \{2, 4, 6\}$$

$$P(E) = \frac{3}{6} = \frac{1}{2}$$

$$P(E) = \underline{\underline{\frac{1}{2}}}$$

ii) An odd number.

$$P(O) = \{1, 3, 5\}$$

$$P(O) = \frac{3}{6} = \underline{\underline{\frac{1}{2}}}$$

$$P(O) = \underline{\underline{\frac{1}{2}}}$$

iii) A number greater than 3.

$$P(>3) = \{4, 5, 6\}$$

$$P(>3) = \frac{3}{6} = \frac{1}{2}$$

$$P(>3) = \underline{\underline{\frac{1}{2}}}$$

Compound Events

A compound event is a probability of 2 or more events happening at once.

The events ~~are~~ can be measured using the following :-

i) Organised Lists.

ii) Tree diagrams.

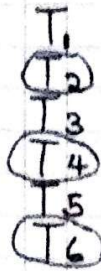
* Use of Organised Lists.

Exercise

1. When you toss a coin and roll a die, what is the probability of getting a tail and even number?

use
Organised Lists:

H₁
H₂
H₃
H₄
H₅
H₆



Total possible outcomes = 12
Tail and even = {T₂, T₄, T₆}

$$P(\text{Tail \& Even}) = \frac{3}{12} \text{ or } \frac{1}{4}$$

2. What is the probability of getting a tail and an odd number?

use
Organised Lists

H₁
H₂
H₃
H₄
H₅
H₆



Total outcomes = 12
Tail and even = {T₁, T₃, T₅}

$$P(\text{Tail \& Odd}) = \frac{3}{12} \text{ or } \frac{1}{4}$$

3. What is the probability of getting each outcome?

$$P(\text{of getting each outcome}) = \left\{ \frac{1}{12} \right\}$$

$$P(E) = \frac{1}{12}$$

4. What is the probability of getting Head and Odd number.

Organised Lists

H_1
 H_2
 H_3
 H_4
 H_5
 H_6

T_1
 T_2
 T_3
 T_4
 T_5
 T_6

Total possible outcomes = 12

$$P(H \& \text{Odd}) = \frac{3}{12}$$

$$P(H \& \text{Odd}) = \{H_1, H_3, H_5\}$$

$$= P(H \& O) = \frac{3}{12} \text{ or } \frac{1}{4}$$

5. What is the probability of getting Head and a number < 2 (less than 2).

Organised Lists

H_1
 H_2
 H_3
 H_4
 H_5
 H_6

T_1
 T_2
 T_3
 T_4
 T_5
 T_6

Total possible outcomes = 12

$$P(H \& < 2) = \{H_1, H_2\}$$

$$P(H \& < 2) = \frac{2}{12} = \frac{1}{6}$$

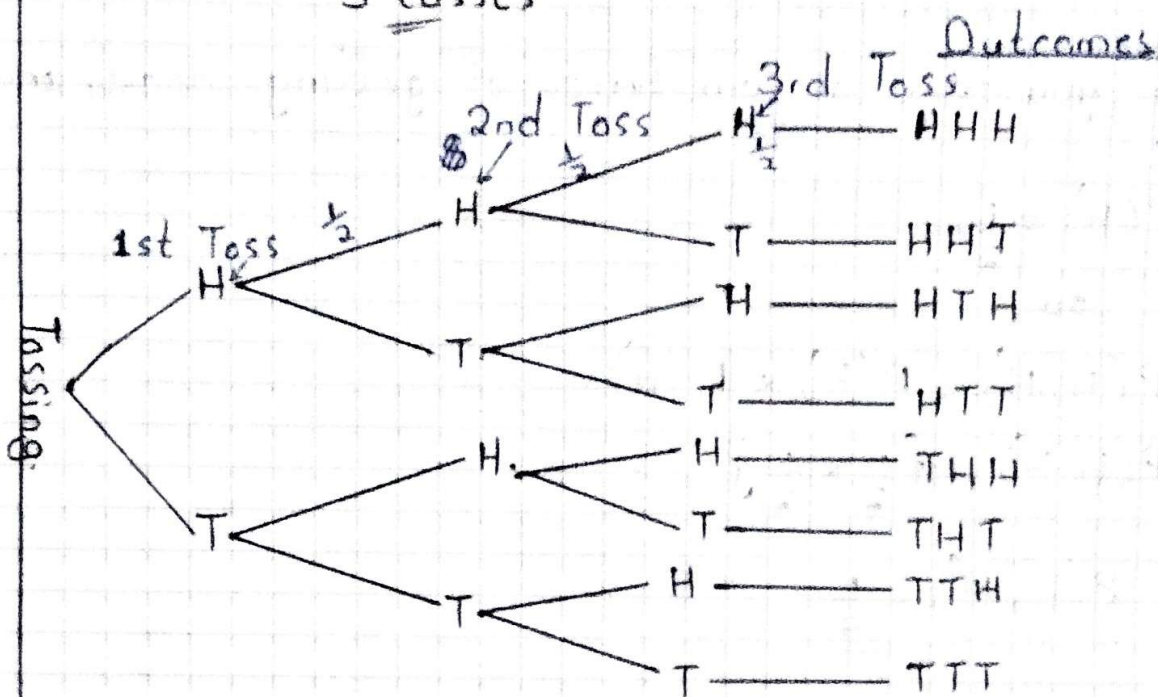
$$P(H \& < 2) = \frac{1}{6}$$

* Use of Tree Diagrams.

1. Exercise.

1. When you toss a coin thrice, find the probability of getting Head, Head, Head.

Use Tree diagram
3 tosses



$$P(H, H, H) = \left\{ \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \right\}$$

$$P(H, H, H) = \frac{1}{8}$$

Addition of Probability.

When we use the word OR we use addition.

$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C).$$

Exercise

1. $P(5 \text{ or } 1)$

$$P(5 \text{ or } 1) = P(5) + P(1)$$

T = 6 outcomes.

number of outcomes (5 & 1) $\Rightarrow 2$

$$P(5 \text{ or } 1) = \frac{2}{6} = \frac{1}{3}$$

$$P(5 \text{ or } 1) = \frac{1}{3}$$

Product of Probability.

When we use AND we use multiplication.

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example

1. When you toss a coin twice of getting Head, and Head.

$$P(H) = \frac{1}{2}$$

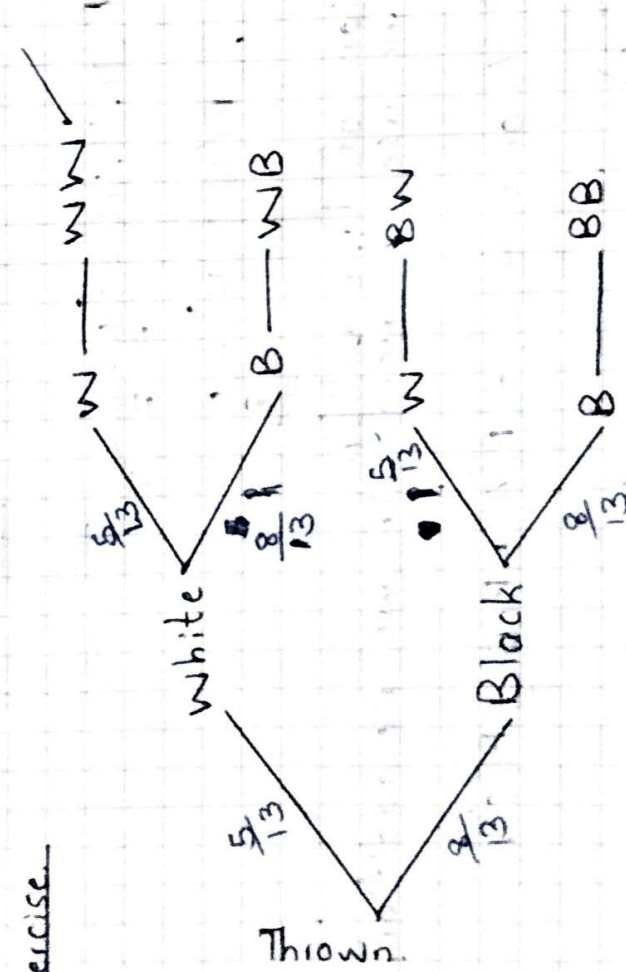
so

$$P(H, H) = P(H) \times P(H)$$

$$P(H, H) = \left\{ \frac{1}{2} \times \frac{1}{2} \right\}$$

$$P(H, H) = \frac{1}{4}$$

Exercise



$$P(W) = \{W \text{ and } W\}$$

$$P(W) = \left(\frac{5}{13} \right) \times \left(\frac{5}{13} \right) = \frac{25}{169}$$

$$P(W) = \frac{25}{169}$$

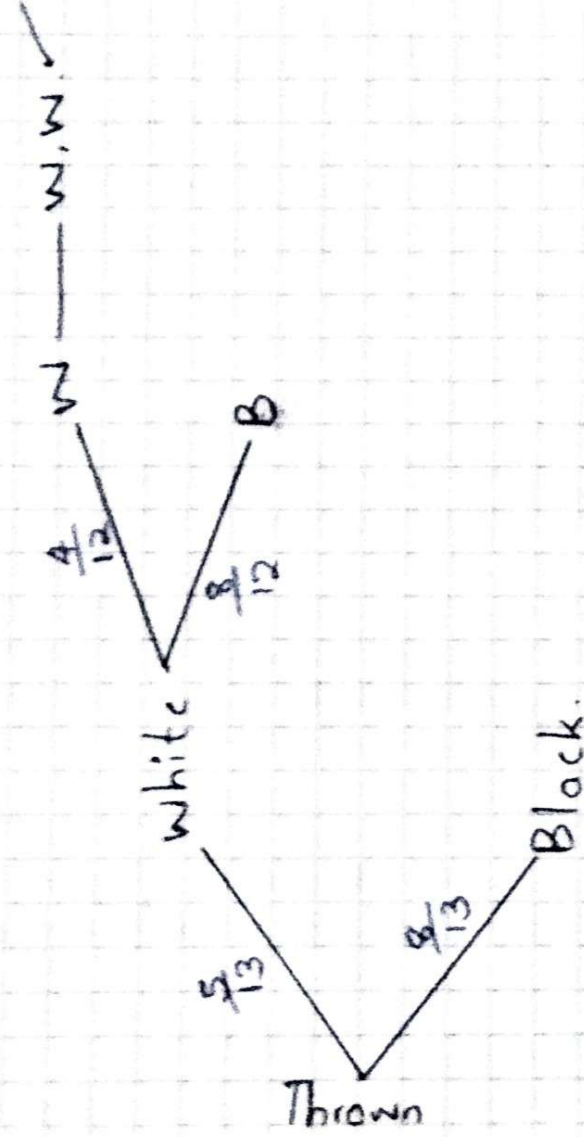
The question was

A tin contains 5 white balls and 8 black balls. Two balls are thrown at random. Find the probability of getting white, white ball. If the white ball is replaced. = $\frac{25}{169}$.

Then 2.

Probability of getting both ball white (w,w) if the balls are not replaced. = $\frac{5}{39}$

$$\frac{5}{13} \times \frac{4}{12} = \frac{4}{12}$$



$$P(w,w) \text{ not replacement} = \left\{ \frac{5}{13} \times \frac{4}{12} \right\} = \frac{20}{156} = \frac{5}{39}$$

$$P(w,w) = \frac{5}{39}$$